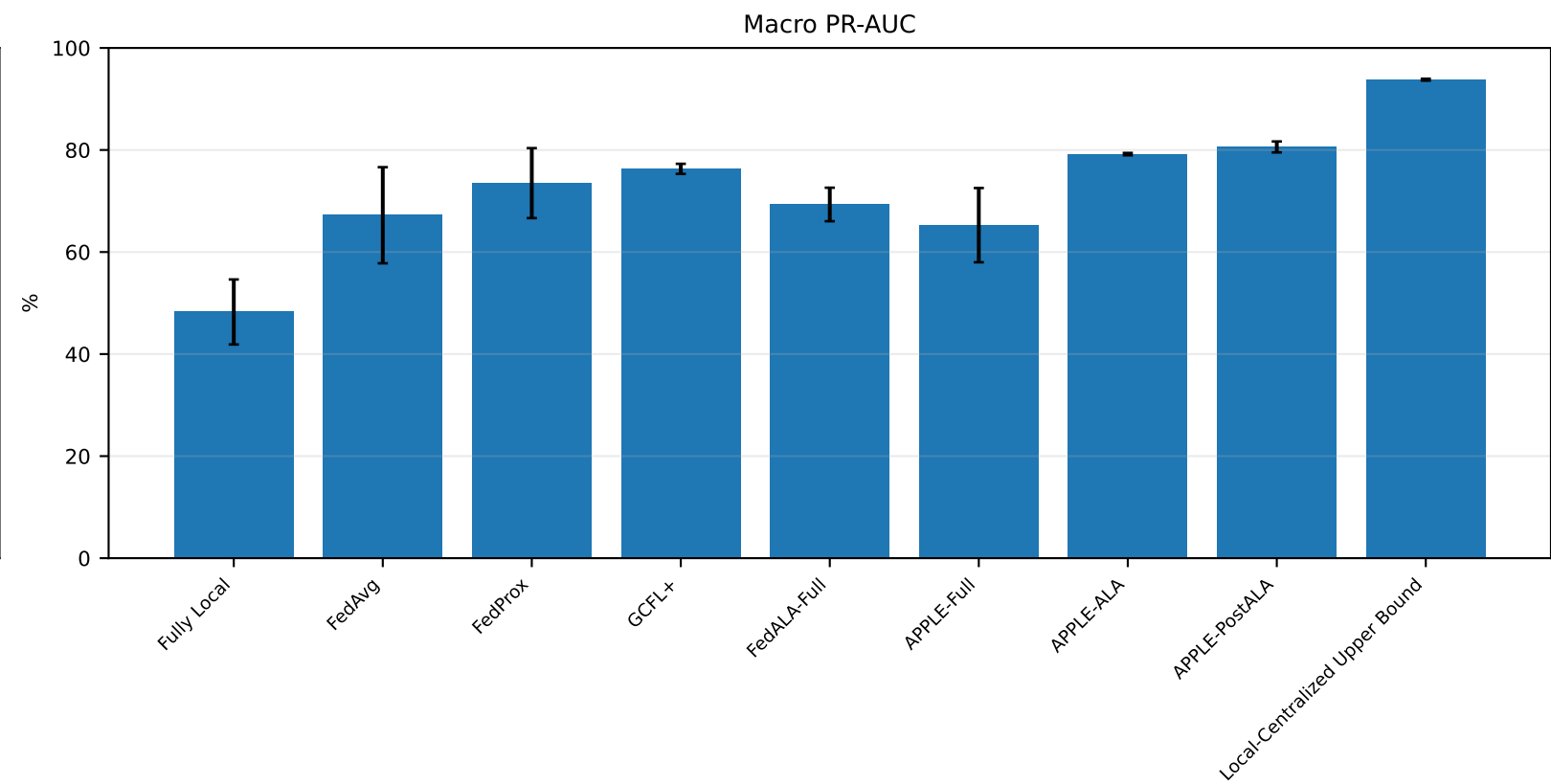
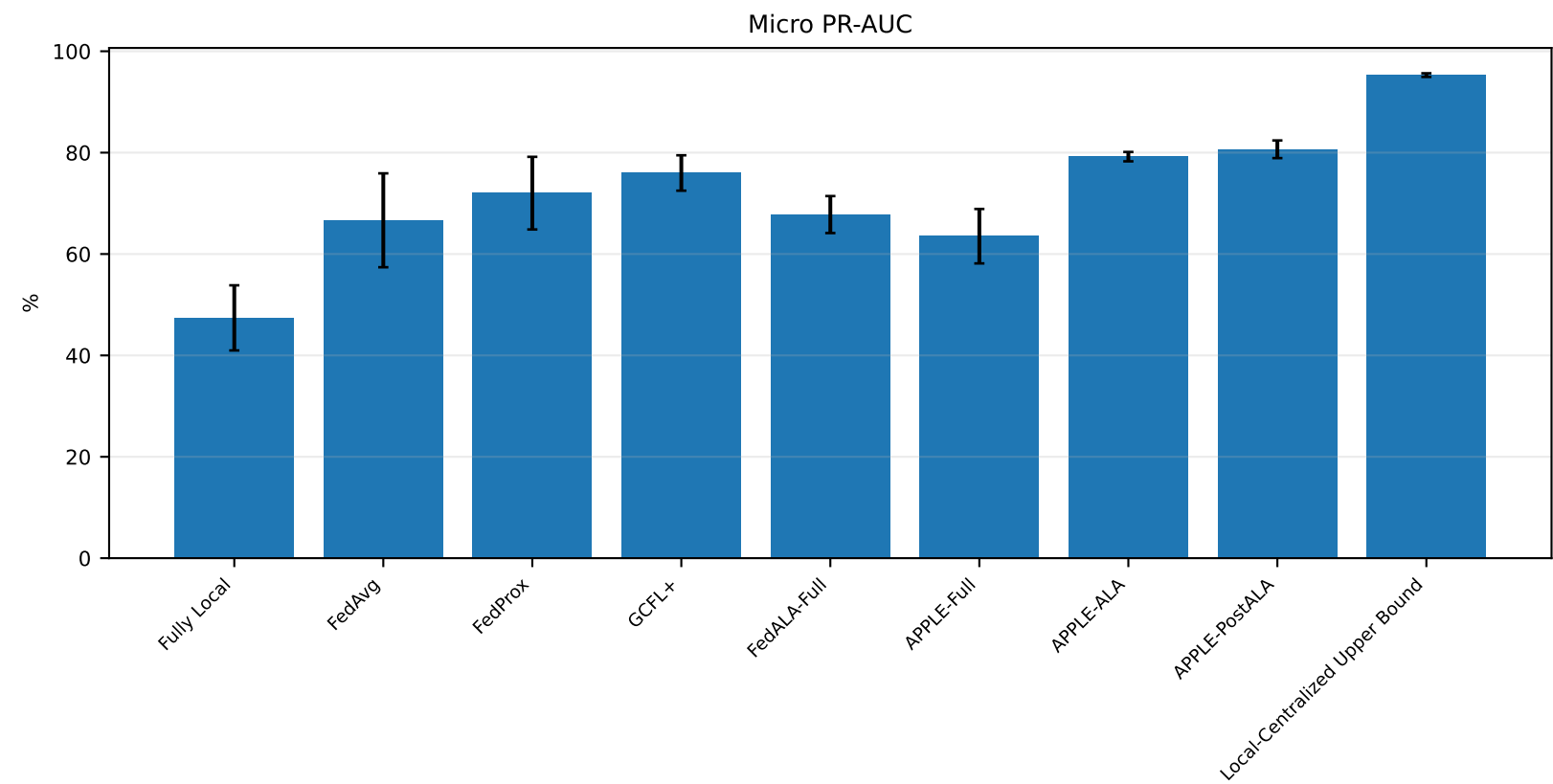
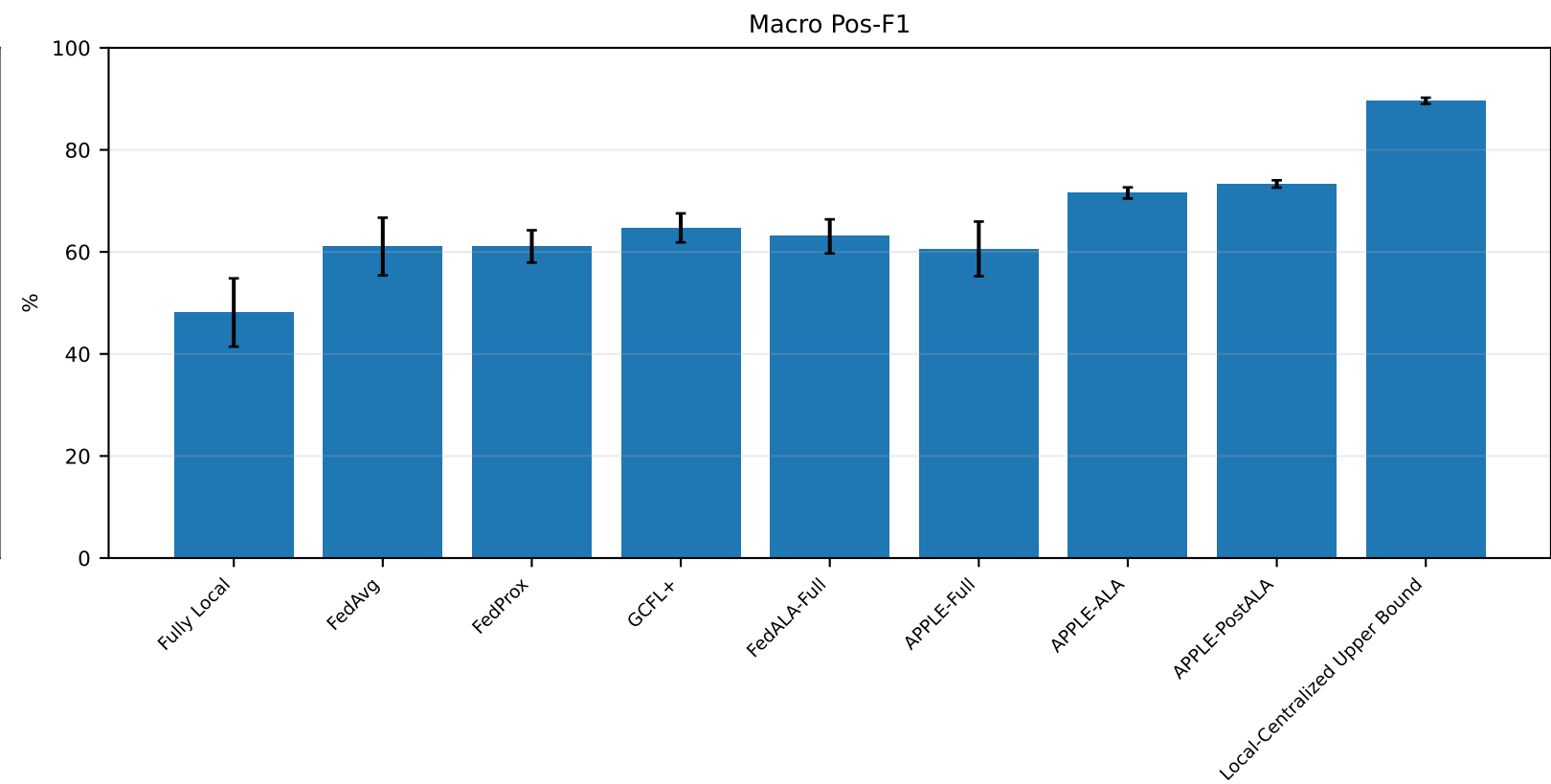
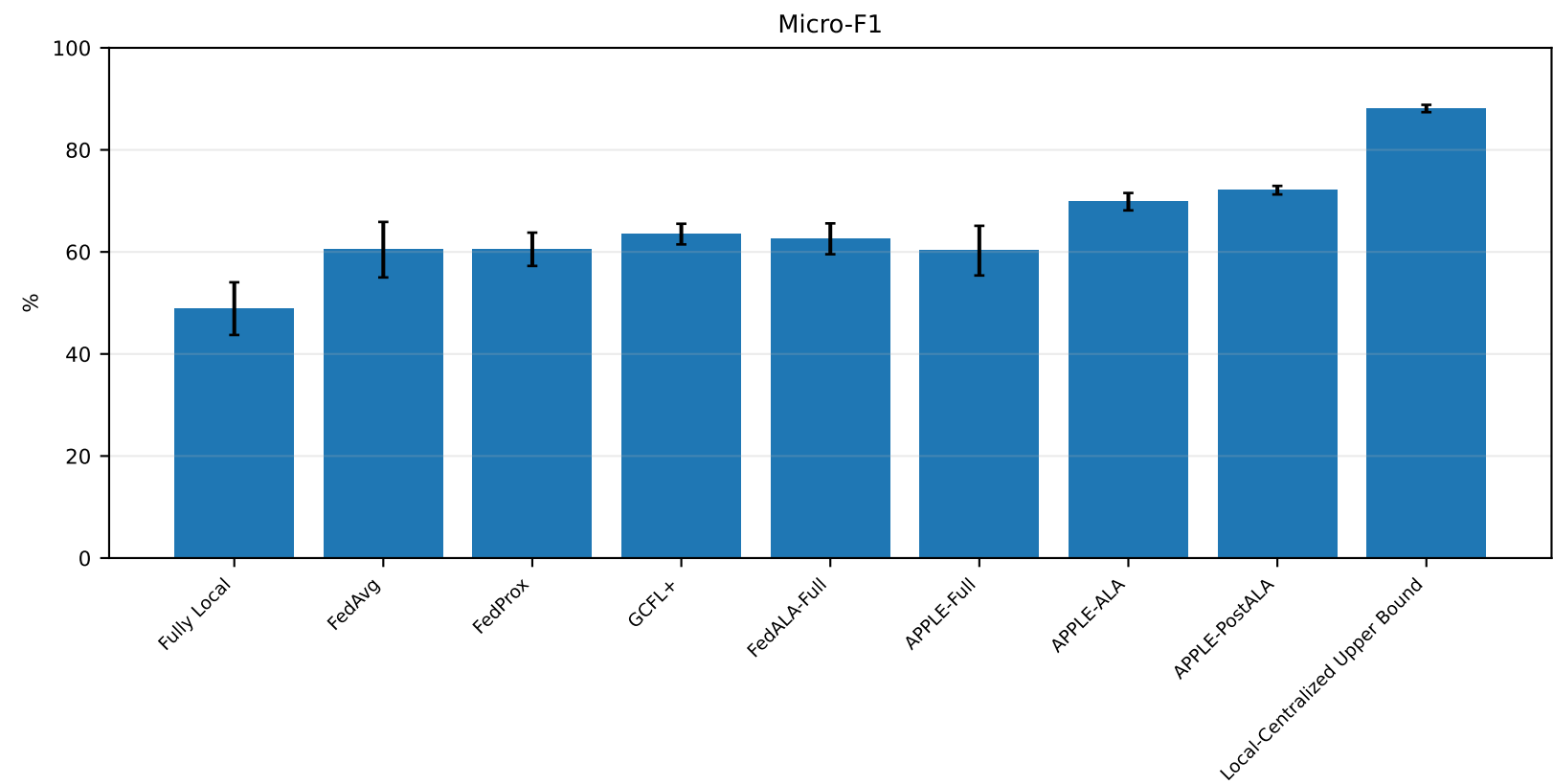


Main baselines + APPLE/FedALA/PostALA | High heterogeneity (q=0.8) | selector=micro_f1 | protocol=OO
O+O: oracle/full validation selection → oracle/full test

Method	Micro-F1	Macro Pos-F1	Micro PR-AUC	Macro PR-AUC	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1	Sel. round	n
Fully Local	48.9% ± 5.2%	48.1% ± 6.7%	47.4% ± 6.4%	48.3% ± 6.4%	51.3% ± 11.2%	51.4% ± 11.0%	50.1% ± 6.8%	47.1% ± 6.4%	40.7% ± 2.2%	—	3
FedAvg	60.4% ± 5.4%	61.0% ± 5.7%	66.7% ± 9.3%	67.2% ± 9.4%	60.7% ± 5.6%	66.8% ± 10.3%	62.6% ± 1.7%	61.7% ± 7.0%	53.4% ± 4.0%	74.7 ± 4.6	3
FedProx	60.5% ± 3.3%	61.1% ± 3.2%	72.0% ± 7.2%	73.5% ± 6.9%	59.4% ± 5.8%	66.8% ± 3.6%	62.9% ± 1.4%	62.5% ± 5.3%	53.9% ± 3.8%	71.7 ± 3.8	3
GCFL+	63.5% ± 2.0%	64.7% ± 2.8%	76.0% ± 3.5%	76.3% ± 1.0%	68.9% ± 11.5%	71.4% ± 4.7%	65.0% ± 3.5%	63.7% ± 1.6%	54.6% ± 1.3%	72.0 ± 12.2	3
FedALA-Full	62.6% ± 3.0%	63.0% ± 3.3%	67.8% ± 3.6%	69.3% ± 3.3%	64.7% ± 5.4%	68.8% ± 6.4%	63.0% ± 0.9%	63.2% ± 3.1%	55.5% ± 1.6%	78.3 ± 2.1	3
APPLE-Full	60.3% ± 4.9%	60.6% ± 5.4%	63.5% ± 5.4%	65.3% ± 7.3%	63.1% ± 12.0%	62.3% ± 6.6%	59.0% ± 3.6%	63.7% ± 5.3%	55.0% ± 2.3%	80.0 ± 0.0	3
APPLE-ALA	69.8% ± 1.7%	71.6% ± 1.1%	79.2% ± 0.9%	79.2% ± 0.2%	81.3% ± 5.5%	78.8% ± 5.6%	70.2% ± 1.7%	70.3% ± 1.2%	57.1% ± 4.3%	72.3 ± 0.6	3
APPLE-PostALA	72.1% ± 0.8%	73.3% ± 0.7%	80.7% ± 1.7%	80.6% ± 1.1%	82.7% ± 0.7%	80.9% ± 1.0%	71.4% ± 2.0%	71.9% ± 1.8%	59.7% ± 2.9%	78.0 ± 2.0	3
Local-Centralized Upper Bound	88.1% ± 0.7%	89.6% ± 0.6%	95.3% ± 0.4%	93.8% ± 0.2%	98.5% ± 0.3%	95.8% ± 0.9%	92.5% ± 1.8%	85.8% ± 0.6%	75.5% ± 2.0%	—	3

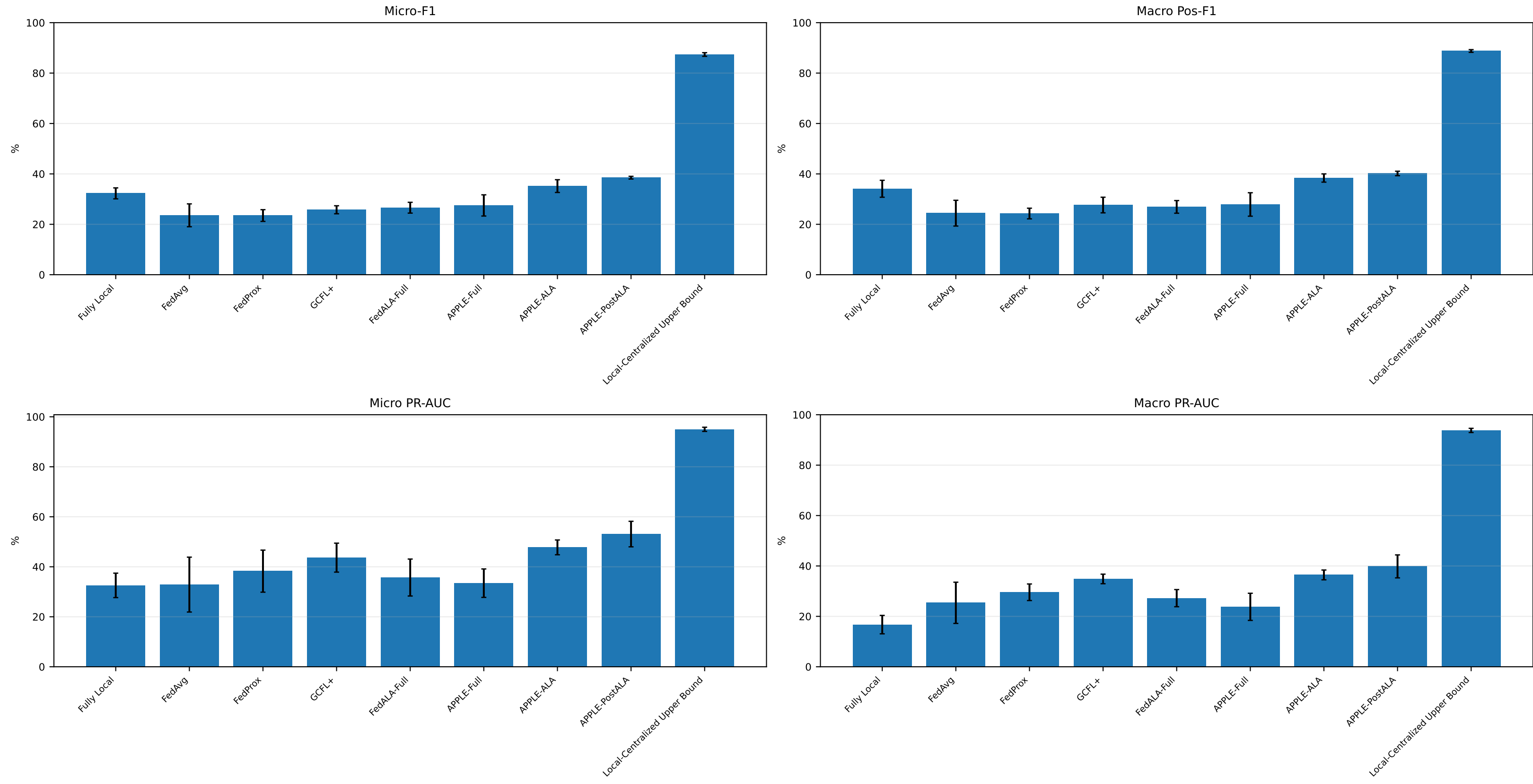
Main baselines + APPLE/FedALA/PostALA | High heterogeneity ($q=0.8$) | selector=micro_f1 | protocol=OO | overview bars
O+O: oracle/full validation selection → oracle/full test



Main baselines + APPLE/FedALA/PostALA | High heterogeneity (q=0.8) | selector=micro_f1 | protocol=VV
V+V: visible validation selection → visible test

Method	Micro-F1	Macro Pos-F1	Micro PR-AUC	Macro PR-AUC	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1	Sel. round	n
Fully Local	32.3% ± 2.2%	34.1% ± 3.4%	32.6% ± 4.9%	16.7% ± 3.7%	48.5% ± 5.8%	39.8% ± 10.2%	28.0% ± 4.2%	30.8% ± 7.2%	23.5% ± 3.8%	—	3
FedAvg	23.6% ± 4.5%	24.4% ± 5.1%	32.9% ± 11.0%	25.4% ± 8.2%	24.1% ± 4.8%	29.6% ± 10.0%	24.9% ± 1.8%	24.7% ± 6.4%	18.8% ± 2.8%	74.0 ± 5.3	3
FedProx	23.5% ± 2.3%	24.3% ± 2.1%	38.3% ± 8.4%	29.6% ± 3.3%	24.0% ± 4.8%	28.3% ± 2.2%	24.9% ± 1.5%	25.2% ± 4.8%	19.0% ± 2.4%	74.0 ± 3.6	3
GCFL+	25.8% ± 1.6%	27.7% ± 3.1%	43.7% ± 5.8%	34.9% ± 1.9%	32.6% ± 13.3%	33.6% ± 5.6%	26.9% ± 2.5%	25.6% ± 1.0%	19.6% ± 0.9%	72.0 ± 12.2	3
FedALA-Full	26.6% ± 2.1%	26.9% ± 2.5%	35.7% ± 7.4%	27.2% ± 3.4%	27.6% ± 3.0%	31.6% ± 6.3%	25.6% ± 1.2%	27.8% ± 2.8%	22.1% ± 0.4%	78.0 ± 2.6	3
APPLE-Full	27.5% ± 4.2%	27.9% ± 4.6%	33.4% ± 5.7%	23.8% ± 5.4%	30.8% ± 7.7%	28.9% ± 5.8%	23.9% ± 2.2%	30.6% ± 6.2%	25.2% ± 2.9%	80.0 ± 0.0	3
APPLE-ALA	35.2% ± 2.5%	38.4% ± 1.6%	47.8% ± 3.0%	36.4% ± 1.9%	44.8% ± 4.3%	51.0% ± 6.3%	36.0% ± 3.6%	35.9% ± 2.5%	24.2% ± 4.6%	72.7 ± 0.6	3
APPLE-PostALA	38.5% ± 0.5%	40.2% ± 0.9%	53.1% ± 5.1%	39.8% ± 4.6%	51.8% ± 1.2%	51.2% ± 3.9%	34.5% ± 6.3%	36.4% ± 2.3%	27.0% ± 2.0%	73.3 ± 3.8	3
Local-Centralized Upper Bound	87.4% ± 0.7%	88.8% ± 0.5%	95.0% ± 0.8%	93.8% ± 0.8%	97.1% ± 0.7%	94.6% ± 1.7%	92.1% ± 1.1%	84.3% ± 0.8%	75.7% ± 1.3%	—	3

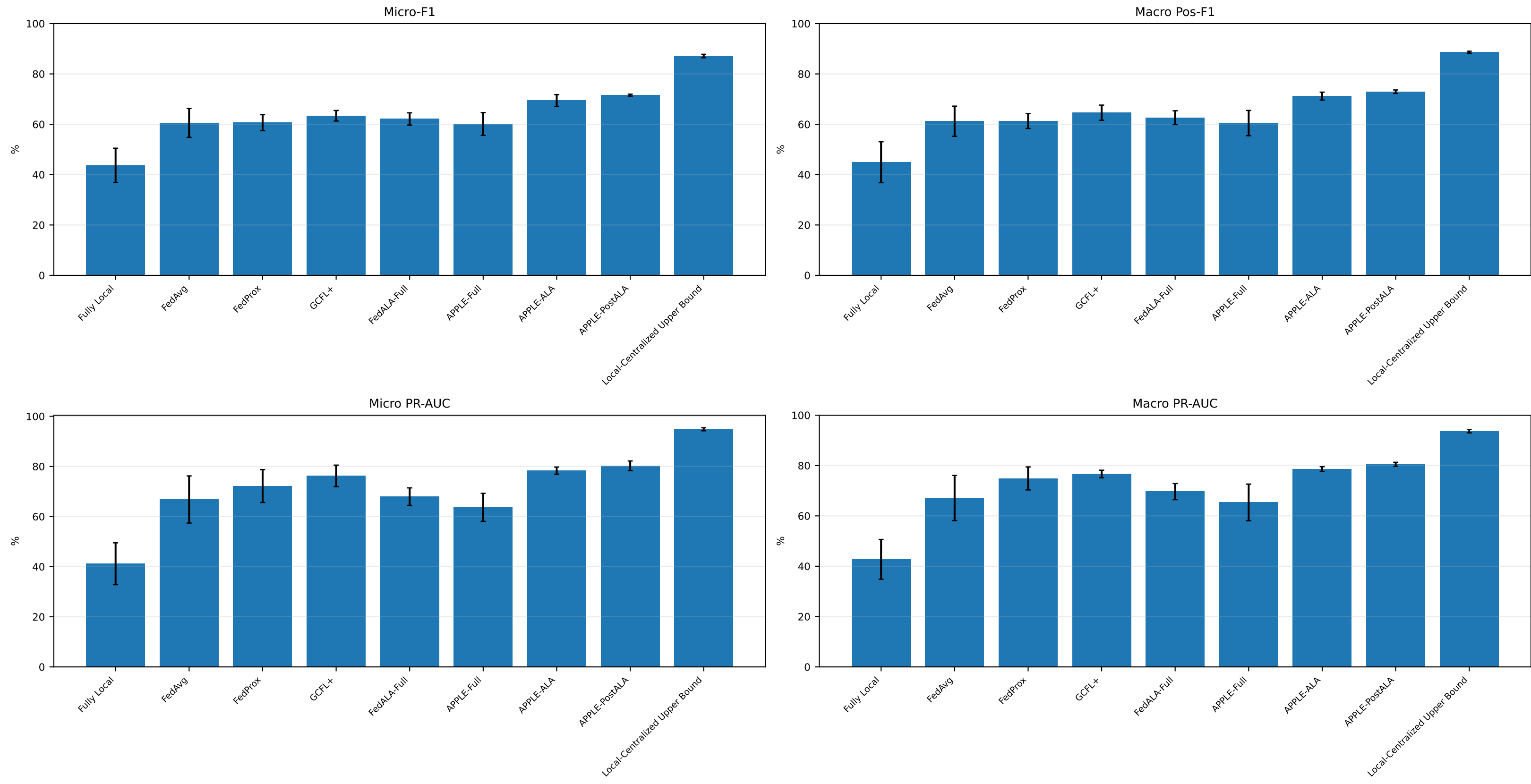
Main baselines + APPLE/FedALA/PostALA | High heterogeneity ($q=0.8$) | selector=micro_f1 | protocol=VV | overview bars
V+V: visible validation selection → visible test



Main baselines + APPLE/FedALA/PostALA | High heterogeneity (q=0.8) | selector=micro_f1 | protocol=VO
V+O: visible validation selection → oracle/full test

Method	Micro-F1	Macro Pos-F1	Micro PR-AUC	Macro PR-AUC	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1	Sel. round	n
Fully Local	43.7% ± 6.8%	44.9% ± 8.1%	41.2% ± 8.3%	42.7% ± 7.9%	50.6% ± 12.2%	48.4% ± 11.1%	45.8% ± 9.9%	43.2% ± 7.9%	36.7% ± 4.7%	—	3
FedAvg	60.5% ± 5.7%	61.2% ± 6.0%	66.8% ± 9.4%	67.1% ± 9.0%	61.4% ± 6.1%	67.1% ± 10.6%	62.5% ± 2.0%	61.9% ± 7.8%	53.2% ± 4.0%	74.0 ± 5.3	3
FedProx	60.6% ± 3.2%	61.3% ± 3.0%	72.2% ± 6.6%	74.9% ± 4.6%	60.8% ± 6.3%	66.7% ± 3.1%	62.4% ± 2.0%	62.7% ± 6.5%	53.9% ± 3.8%	74.0 ± 3.6	3
GCFL+	63.4% ± 2.1%	64.6% ± 3.0%	76.2% ± 4.3%	76.6% ± 1.5%	68.8% ± 12.1%	71.6% ± 5.2%	64.6% ± 3.0%	63.6% ± 1.3%	54.5% ± 1.1%	72.0 ± 12.2	3
FedALA-Full	62.1% ± 2.4%	62.6% ± 2.7%	67.9% ± 3.5%	69.6% ± 3.2%	64.1% ± 4.7%	69.0% ± 6.2%	62.4% ± 0.6%	63.3% ± 2.7%	54.4% ± 1.6%	78.0 ± 2.6	3
APPLE-Full	60.1% ± 4.5%	60.5% ± 5.0%	63.7% ± 5.6%	65.4% ± 7.3%	63.0% ± 11.1%	62.3% ± 6.5%	58.8% ± 3.0%	63.5% ± 5.8%	54.8% ± 2.2%	80.0 ± 0.0	3
APPLE-ALA	69.4% ± 2.3%	71.2% ± 1.6%	78.3% ± 1.4%	78.6% ± 0.9%	79.3% ± 2.8%	79.1% ± 4.8%	71.2% ± 3.5%	69.6% ± 3.4%	56.7% ± 5.1%	72.7 ± 0.6	3
APPLE-PostALA	71.6% ± 0.4%	72.9% ± 0.7%	80.2% ± 1.9%	80.5% ± 0.8%	83.0% ± 0.8%	82.5% ± 2.6%	68.9% ± 4.7%	71.5% ± 1.5%	58.8% ± 1.9%	73.3 ± 3.8	3
Local-Centralized Upper Bound	87.1% ± 0.7%	88.6% ± 0.4%	94.8% ± 0.6%	93.6% ± 0.7%	96.9% ± 1.2%	95.6% ± 1.4%	91.4% ± 1.8%	84.6% ± 1.7%	74.7% ± 1.3%	—	3

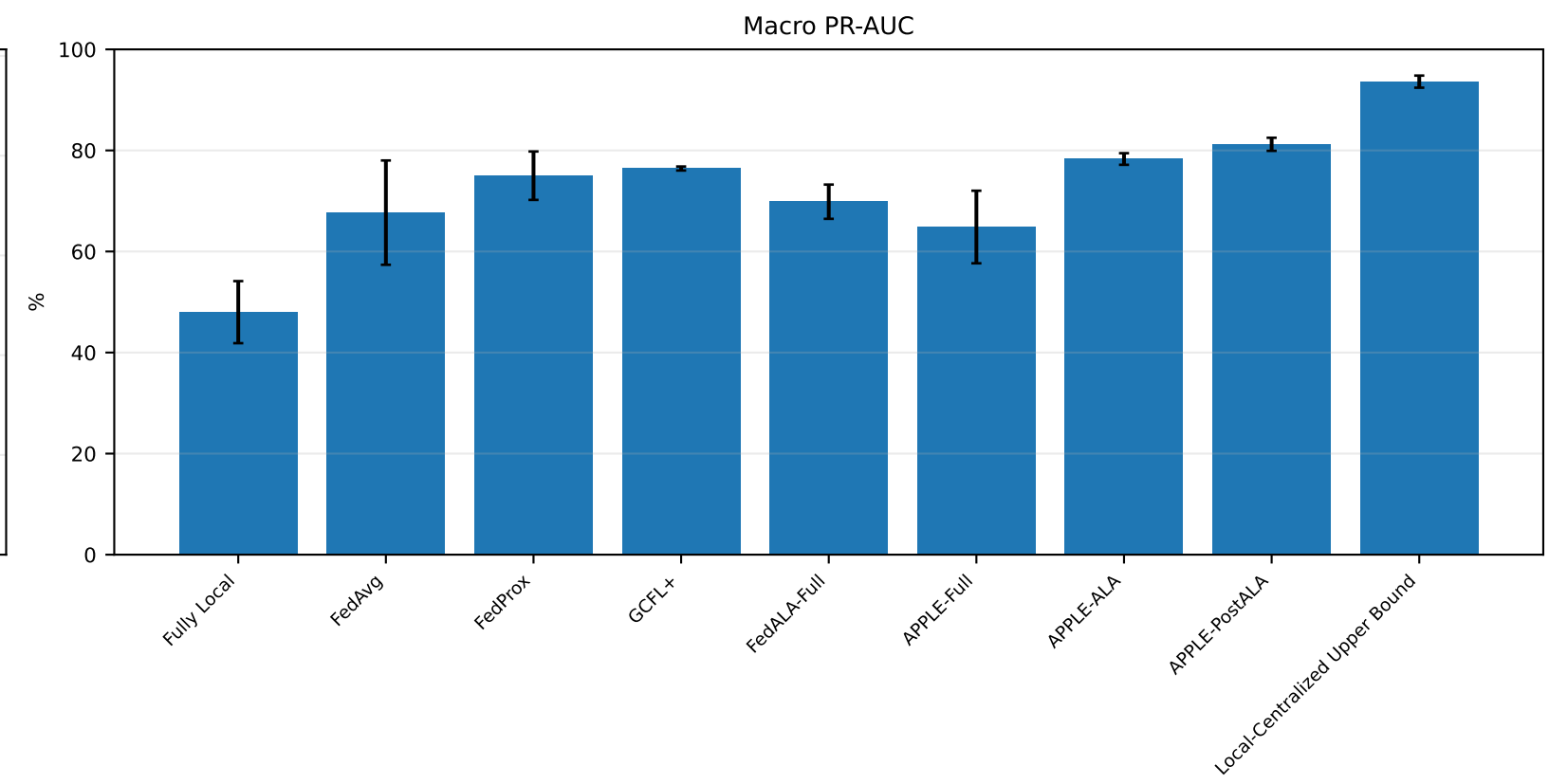
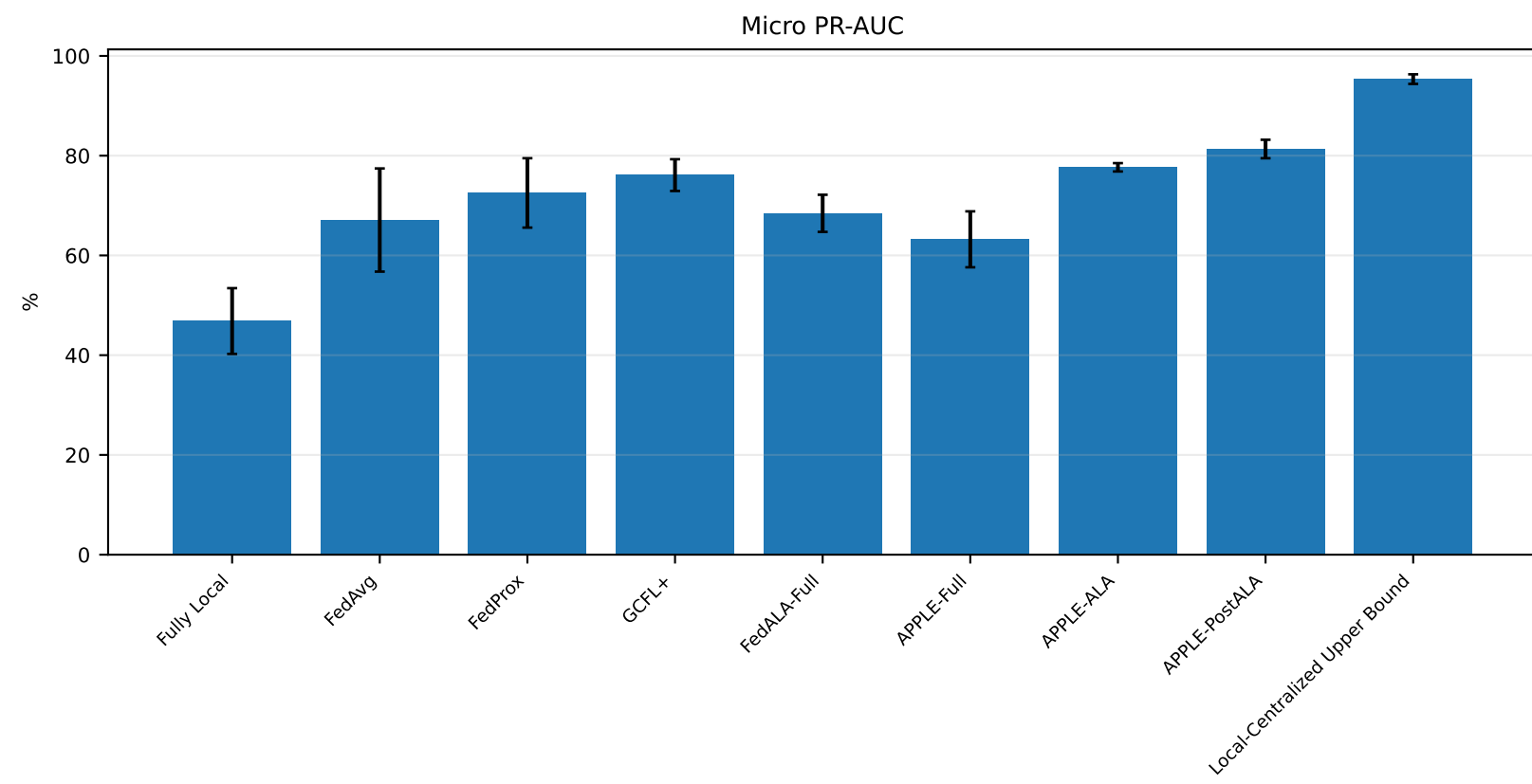
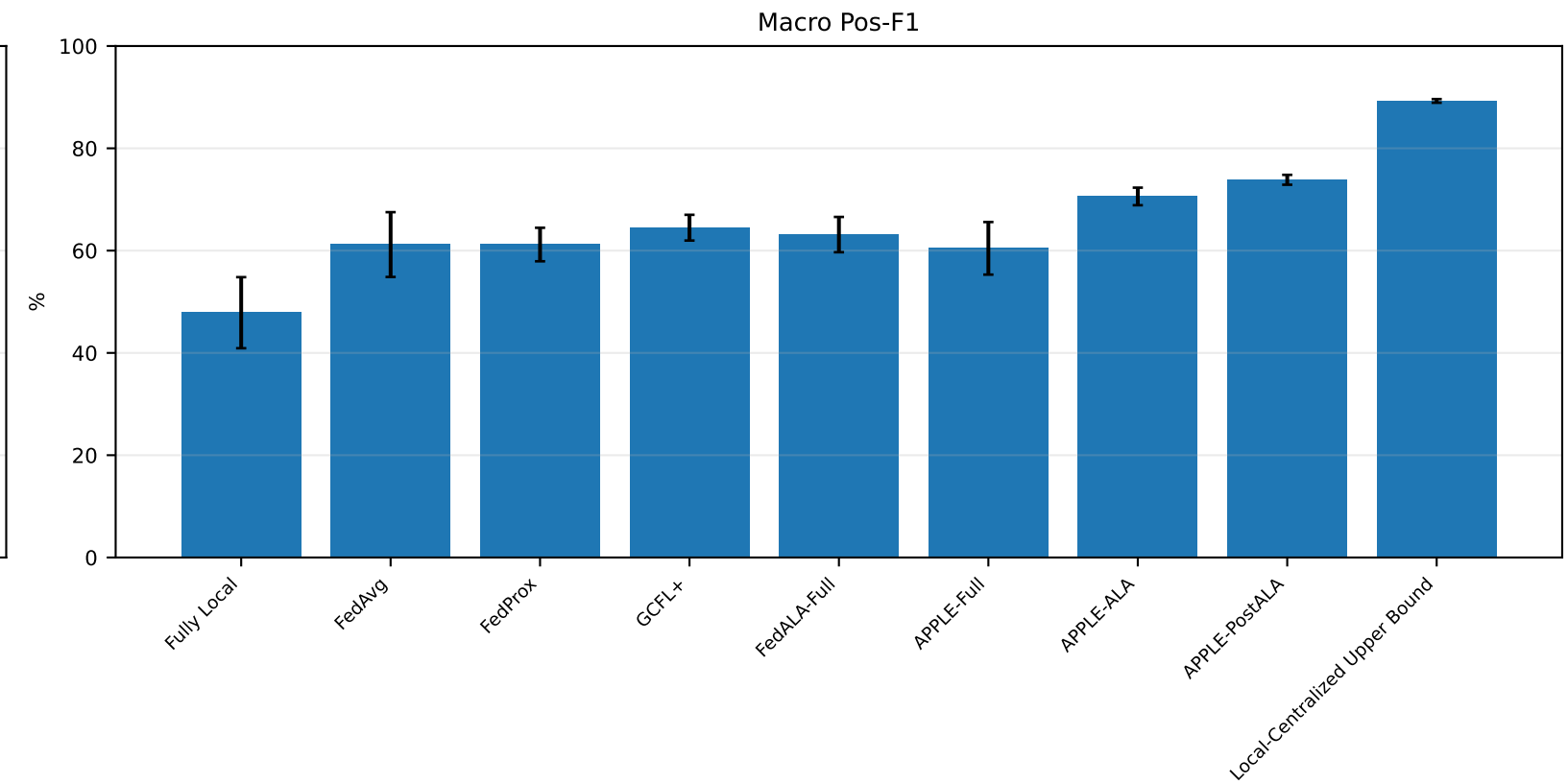
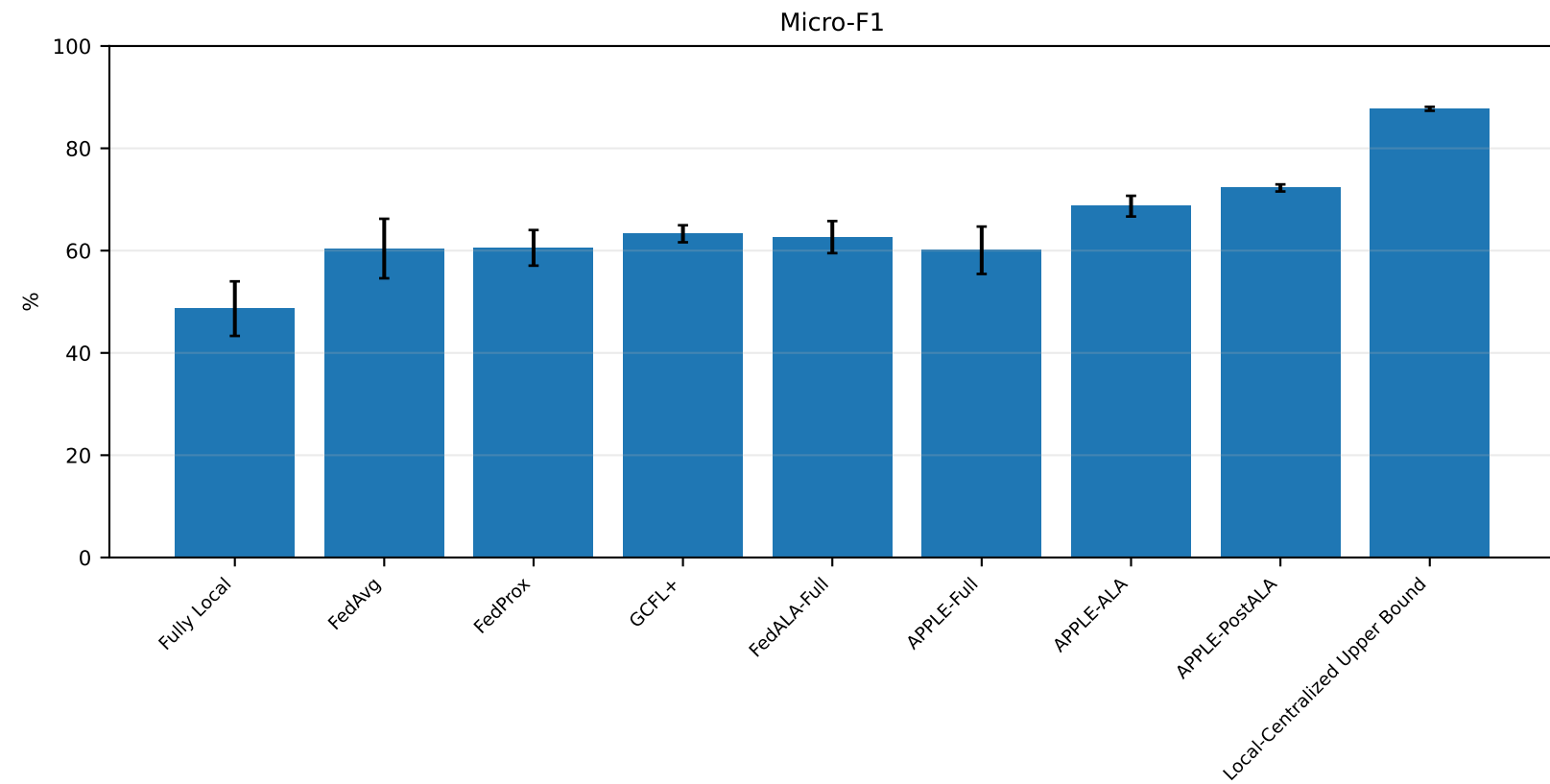
Main baselines + APPLE/FedALA/PostALA | High heterogeneity ($q=0.8$) | selector=micro_f1 | protocol=VO | overview bars
V+O: visible validation selection → oracle/full test



Main baselines + APPLE/FedALA/PostALA | High heterogeneity (q=0.8) | selector=macro_pos_f1 | protocol=OO
O+O: oracle/full validation selection → oracle/full test

Method	Micro-F1	Macro Pos-F1	Micro PR-AUC	Macro PR-AUC	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1	Sel. round	n
Fully Local	48.7% ± 5.3%	47.9% ± 6.9%	46.8% ± 6.6%	48.0% ± 6.1%	51.6% ± 11.5%	50.9% ± 11.5%	49.4% ± 7.3%	46.7% ± 6.6%	40.7% ± 3.0%	—	3
FedAvg	60.4% ± 5.8%	61.2% ± 6.3%	67.1% ± 10.3%	67.7% ± 10.3%	63.0% ± 9.9%	66.2% ± 10.0%	62.5% ± 2.3%	61.3% ± 6.4%	52.9% ± 3.4%	75.0 ± 4.4	3
FedProx	60.5% ± 3.5%	61.2% ± 3.3%	72.5% ± 7.0%	75.0% ± 4.8%	60.5% ± 6.7%	67.0% ± 3.2%	62.3% ± 2.2%	62.3% ± 6.1%	53.9% ± 4.2%	73.7 ± 3.2	3
GCFL+	63.3% ± 1.7%	64.5% ± 2.5%	76.1% ± 3.2%	76.4% ± 0.4%	68.7% ± 11.3%	70.9% ± 5.0%	64.8% ± 3.1%	63.5% ± 0.9%	54.5% ± 1.4%	72.0 ± 12.2	3
FedALA-Full	62.6% ± 3.1%	63.1% ± 3.4%	68.4% ± 3.7%	69.9% ± 3.4%	64.7% ± 5.6%	68.8% ± 6.4%	63.1% ± 0.8%	63.5% ± 2.7%	55.5% ± 2.0%	78.3 ± 2.1	3
APPLE-Full	60.1% ± 4.6%	60.4% ± 5.1%	63.2% ± 5.6%	64.8% ± 7.2%	63.2% ± 11.5%	62.1% ± 6.7%	58.3% ± 3.3%	63.7% ± 5.6%	55.0% ± 2.2%	80.0 ± 0.0	3
APPLE-ALA	68.7% ± 2.0%	70.6% ± 1.7%	77.7% ± 0.8%	78.3% ± 1.1%	81.1% ± 3.6%	78.8% ± 4.1%	68.6% ± 1.6%	68.0% ± 5.0%	56.4% ± 3.5%	75.3 ± 2.9	3
APPLE-PostALA	72.3% ± 0.7%	73.8% ± 1.0%	81.3% ± 1.9%	81.2% ± 1.3%	84.5% ± 1.0%	83.0% ± 4.3%	71.5% ± 2.5%	71.4% ± 2.1%	58.8% ± 1.8%	77.7 ± 1.5	3
Local-Centralized Upper Bound	87.7% ± 0.4%	89.3% ± 0.4%	95.4% ± 1.0%	93.6% ± 1.2%	98.4% ± 0.7%	95.5% ± 0.6%	91.9% ± 1.4%	85.1% ± 0.7%	75.3% ± 0.9%	—	3

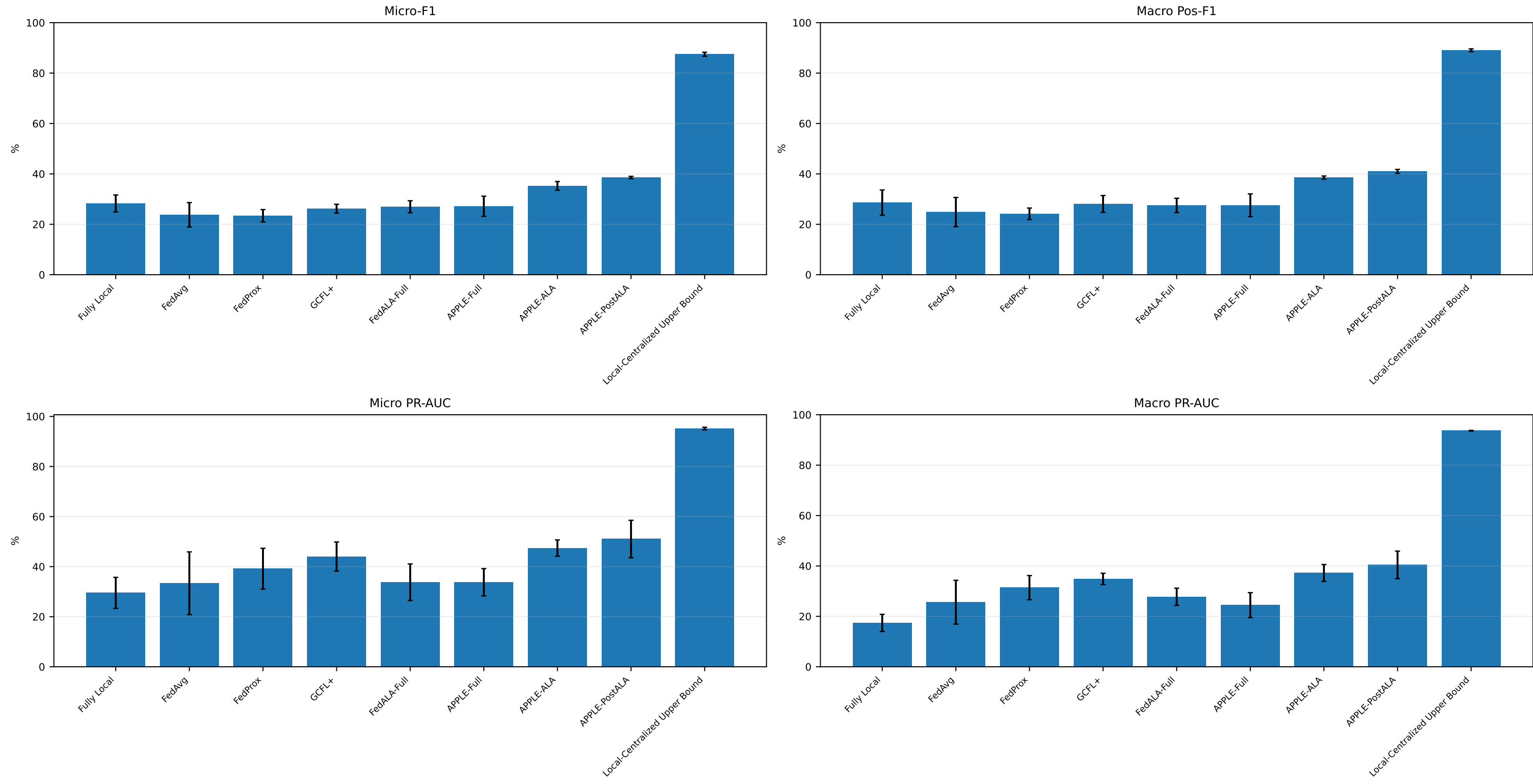
Main baselines + APPLE/FedALA/PostALA | High heterogeneity ($q=0.8$) | selector=macro_pos_f1 | protocol=OO | overview bars
O+O: oracle/full validation selection → oracle/full test



Main baselines + APPLE/FedALA/PostALA | High heterogeneity (q=0.8) | selector=macro_pos_f1 | protocol=VV
V+V: visible validation selection → visible test

Method	Micro-F1	Macro Pos-F1	Micro PR-AUC	Macro PR-AUC	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1	Sel. round	n
Fully Local	28.3% ± 3.4%	28.6% ± 5.0%	29.5% ± 6.2%	17.4% ± 3.4%	36.8% ± 7.6%	33.5% ± 9.5%	24.6% ± 4.7%	27.0% ± 7.4%	21.2% ± 1.5%	—	3
FedAvg	23.8% ± 4.8%	24.9% ± 5.8%	33.4% ± 12.5%	25.6% ± 8.7%	26.6% ± 9.0%	29.6% ± 9.8%	25.2% ± 1.9%	24.4% ± 6.0%	18.5% ± 2.4%	74.3 ± 5.1	3
FedProx	23.4% ± 2.5%	24.2% ± 2.3%	39.2% ± 8.2%	31.4% ± 4.8%	23.6% ± 4.4%	28.4% ± 2.7%	24.8% ± 1.5%	25.1% ± 4.9%	19.0% ± 2.5%	74.0 ± 3.6	3
GCFL+	26.2% ± 1.8%	28.1% ± 3.3%	44.0% ± 5.8%	34.9% ± 2.3%	33.1% ± 14.0%	34.0% ± 4.9%	27.2% ± 2.9%	26.3% ± 1.0%	19.9% ± 0.9%	72.0 ± 12.2	3
FedALA-Full	27.0% ± 2.4%	27.5% ± 2.9%	33.8% ± 7.3%	27.8% ± 3.4%	29.2% ± 3.8%	32.4% ± 6.3%	26.2% ± 1.7%	27.1% ± 2.2%	22.6% ± 0.6%	78.3 ± 2.1	3
APPLE-Full	27.2% ± 4.0%	27.6% ± 4.5%	33.8% ± 5.4%	24.5% ± 4.9%	30.7% ± 7.3%	28.6% ± 5.9%	23.4% ± 2.0%	30.5% ± 5.8%	24.6% ± 3.1%	80.0 ± 0.0	3
APPLE-ALA	35.3% ± 1.7%	38.5% ± 0.6%	47.4% ± 3.3%	37.2% ± 3.3%	48.4% ± 0.2%	47.3% ± 1.4%	37.4% ± 4.5%	35.7% ± 2.0%	23.9% ± 3.6%	71.0 ± 2.6	3
APPLE-PostALA	38.6% ± 0.5%	41.0% ± 0.8%	51.0% ± 7.5%	40.4% ± 5.5%	53.7% ± 3.4%	52.5% ± 4.0%	36.5% ± 4.1%	36.1% ± 2.2%	26.2% ± 2.0%	71.7 ± 3.8	3
Local-Centralized Upper Bound	87.5% ± 0.8%	89.1% ± 0.6%	95.2% ± 0.5%	93.7% ± 0.1%	97.8% ± 1.3%	96.0% ± 0.5%	92.4% ± 1.2%	84.3% ± 1.7%	74.9% ± 1.6%	—	3

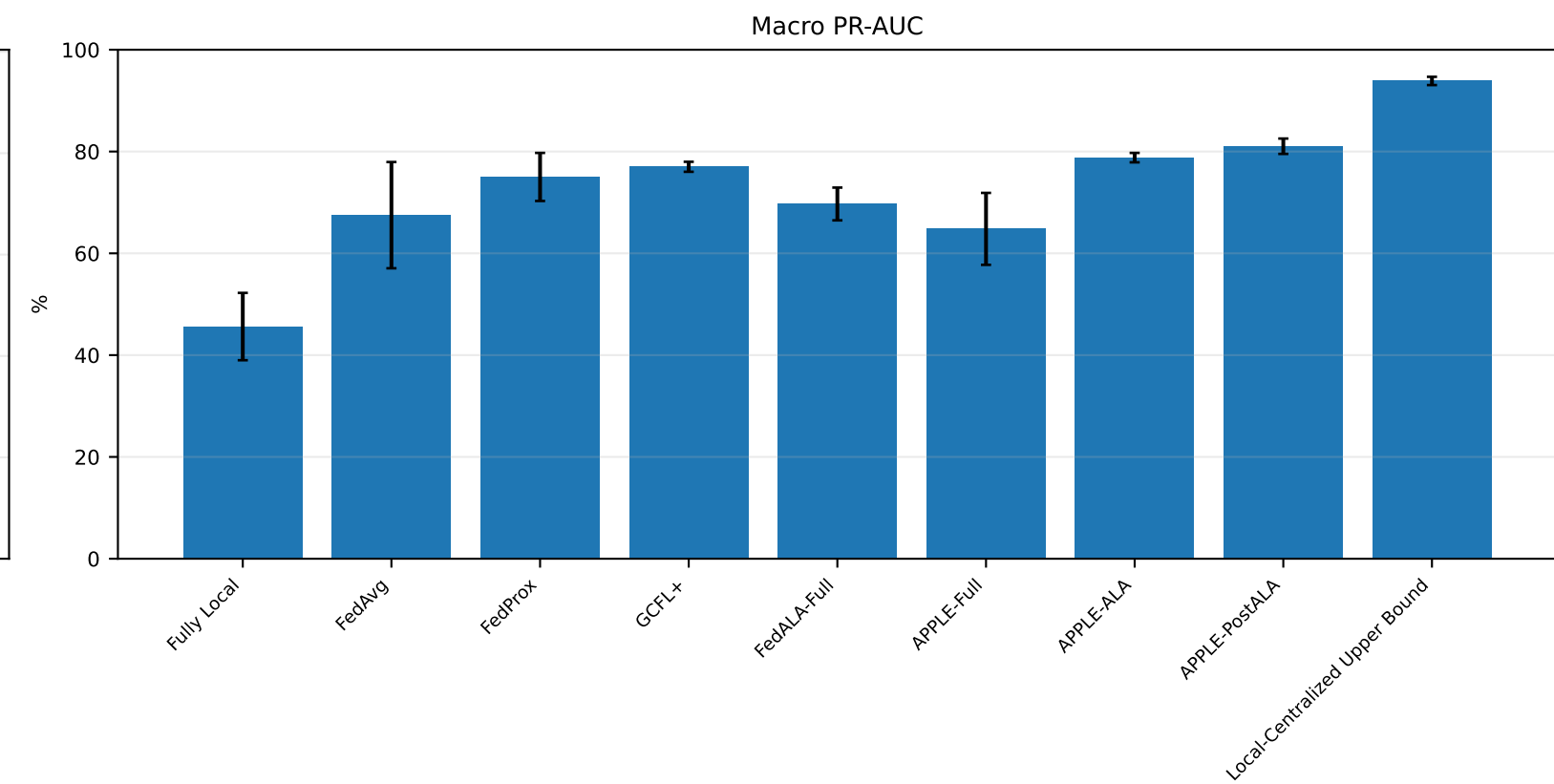
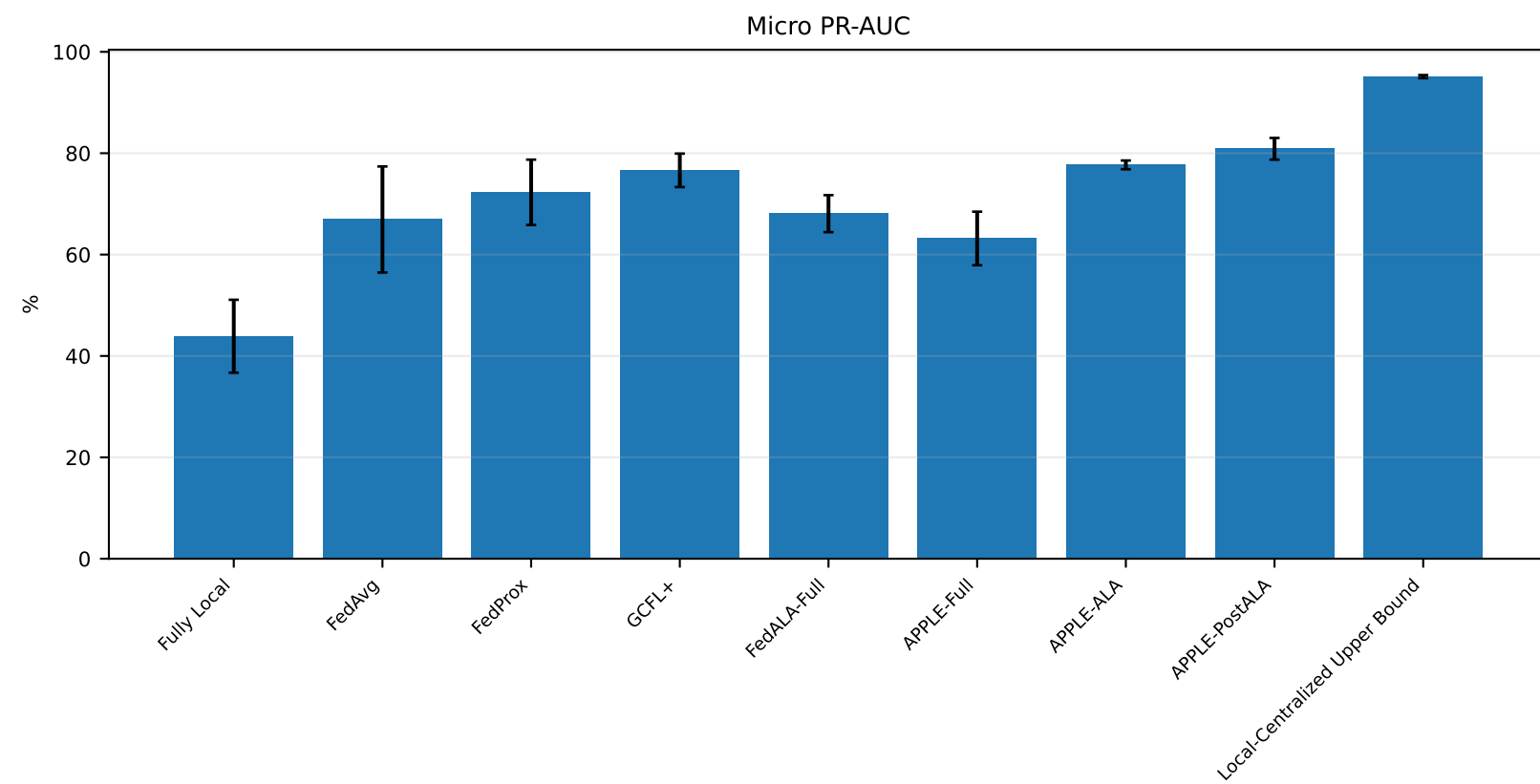
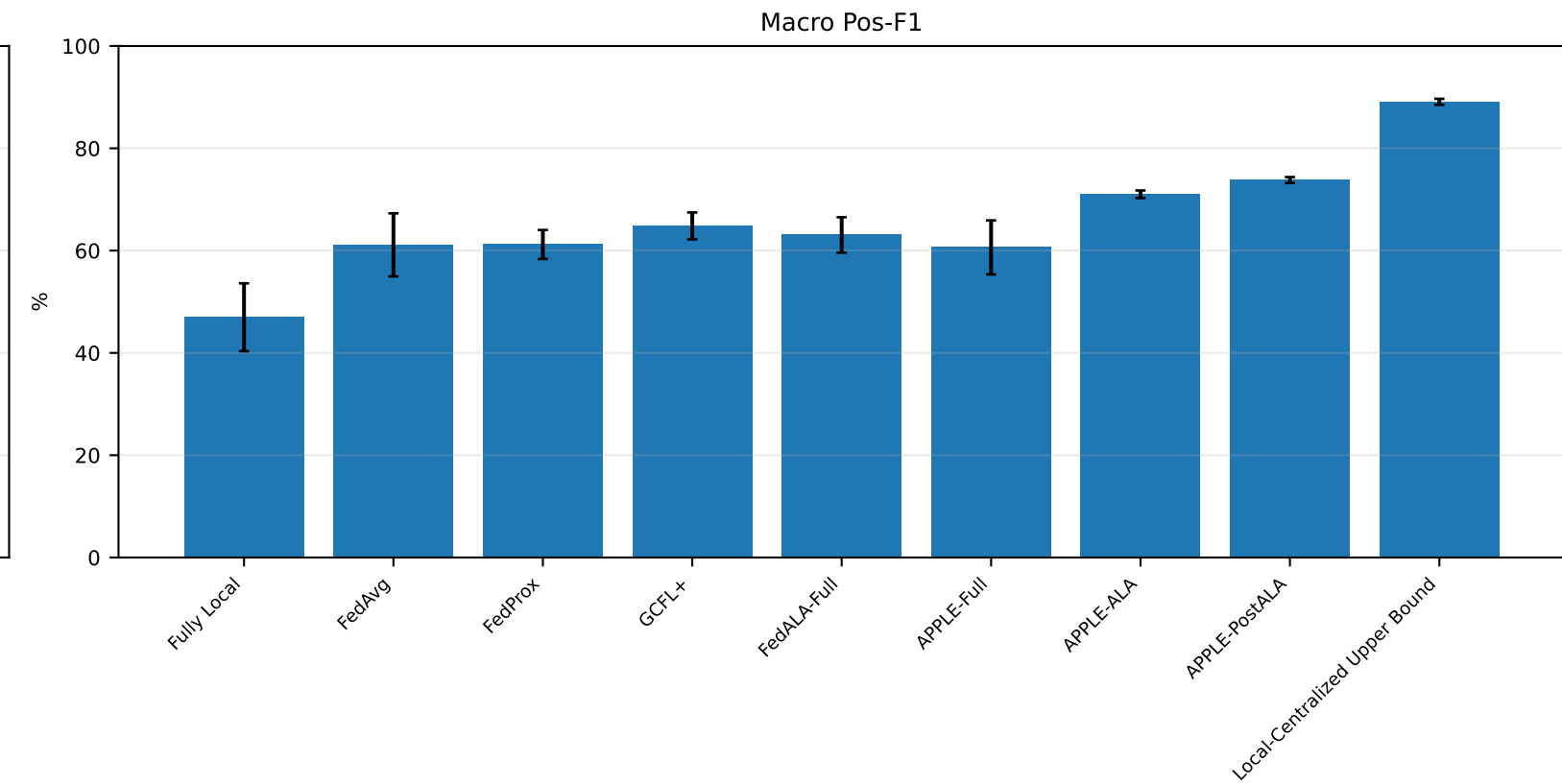
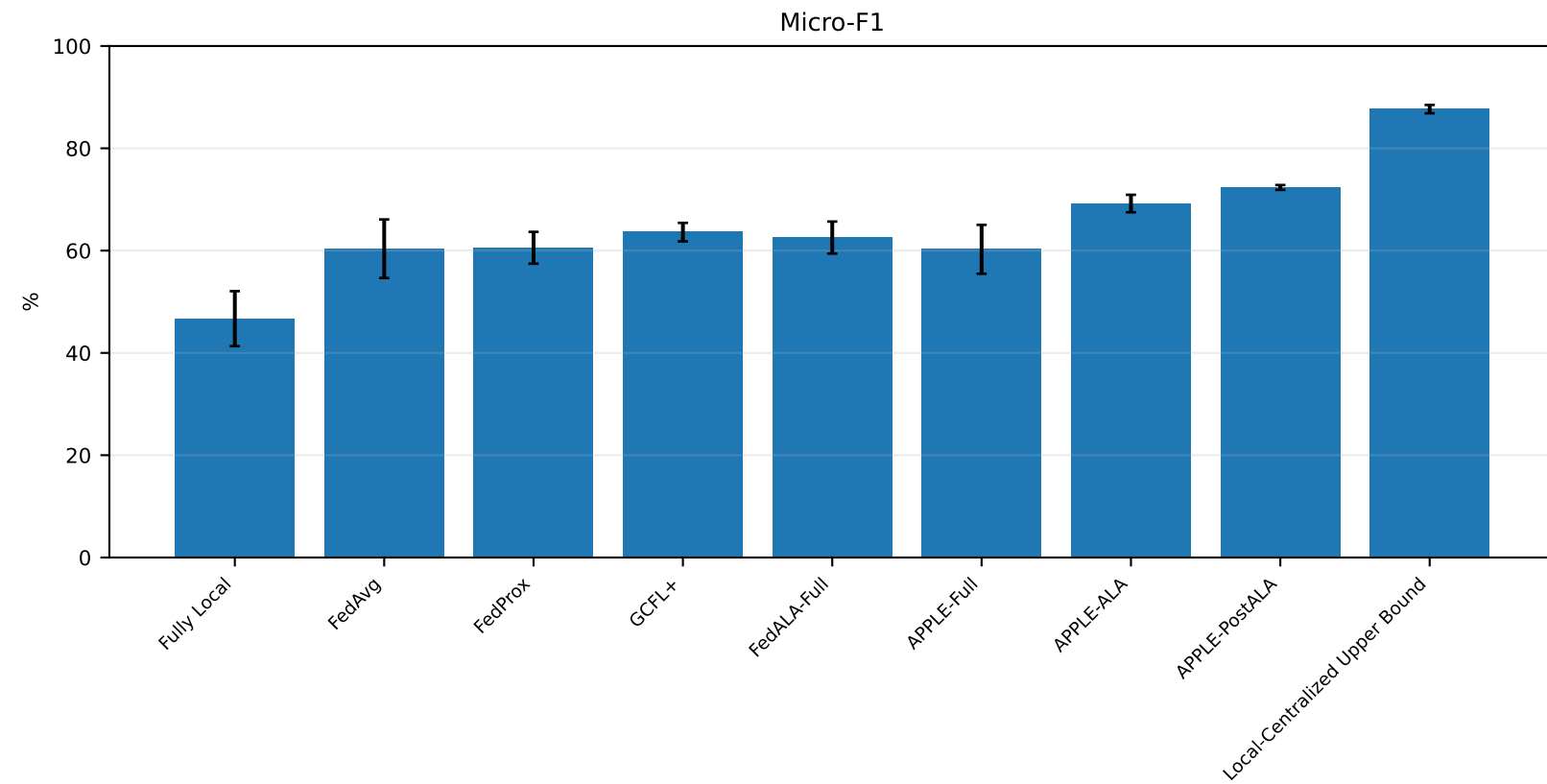
Main baselines + APPLE/FedALA/PostALA | High heterogeneity ($q=0.8$) | selector=macro_pos_f1 | protocol=VV | overview bars
V+V: visible validation selection → visible test



Main baselines + APPLE/FedALA/PostALA | High heterogeneity (q=0.8) | selector=macro_pos_f1 | protocol=VO
V+O: visible validation selection → oracle/full test

Method	Micro-F1	Macro Pos-F1	Micro PR-AUC	Macro PR-AUC	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1	Sel. round	n
Fully Local	46.7% ± 5.4%	47.0% ± 6.6%	43.9% ± 7.2%	45.6% ± 6.6%	51.5% ± 10.9%	50.7% ± 9.6%	47.3% ± 8.0%	45.7% ± 6.6%	39.7% ± 3.3%	—	3
FedAvg	60.4% ± 5.7%	61.1% ± 6.2%	66.9% ± 10.5%	67.5% ± 10.4%	62.7% ± 9.0%	66.4% ± 9.7%	62.4% ± 2.3%	61.2% ± 6.8%	52.9% ± 3.6%	74.3 ± 5.1	3
FedProx	60.6% ± 3.1%	61.2% ± 2.8%	72.3% ± 6.4%	75.0% ± 4.7%	60.6% ± 6.2%	66.4% ± 3.1%	62.3% ± 1.6%	62.7% ± 6.3%	54.0% ± 4.1%	74.0 ± 3.6	3
GCFL+	63.6% ± 1.8%	64.8% ± 2.6%	76.6% ± 3.3%	77.0% ± 1.0%	69.3% ± 11.7%	71.4% ± 4.4%	64.7% ± 3.5%	64.0% ± 1.5%	54.8% ± 1.7%	72.0 ± 12.2	3
FedALA-Full	62.6% ± 3.1%	63.1% ± 3.5%	68.1% ± 3.7%	69.7% ± 3.2%	65.0% ± 5.8%	68.7% ± 6.5%	63.0% ± 0.7%	62.8% ± 3.1%	55.8% ± 1.7%	78.3 ± 2.1	3
APPLE-Full	60.3% ± 4.8%	60.6% ± 5.3%	63.2% ± 5.3%	64.8% ± 7.1%	63.2% ± 11.6%	62.8% ± 7.1%	58.5% ± 3.3%	63.8% ± 5.9%	55.0% ± 2.5%	80.0 ± 0.0	3
APPLE-ALA	69.2% ± 1.7%	71.0% ± 0.7%	77.7% ± 0.9%	78.8% ± 0.9%	79.3% ± 3.6%	78.1% ± 2.3%	71.8% ± 3.9%	69.6% ± 3.4%	56.3% ± 5.2%	71.0 ± 2.6	3
APPLE-PostALA	72.3% ± 0.5%	73.8% ± 0.6%	80.9% ± 2.1%	81.0% ± 1.5%	83.0% ± 2.0%	83.3% ± 2.4%	71.6% ± 3.5%	71.5% ± 1.5%	59.6% ± 1.9%	71.7 ± 3.8	3
Local-Centralized Upper Bound	87.7% ± 0.8%	89.1% ± 0.6%	95.1% ± 0.3%	93.9% ± 0.8%	97.2% ± 1.1%	95.5% ± 1.4%	92.6% ± 1.0%	84.3% ± 2.2%	76.0% ± 1.4%	—	3

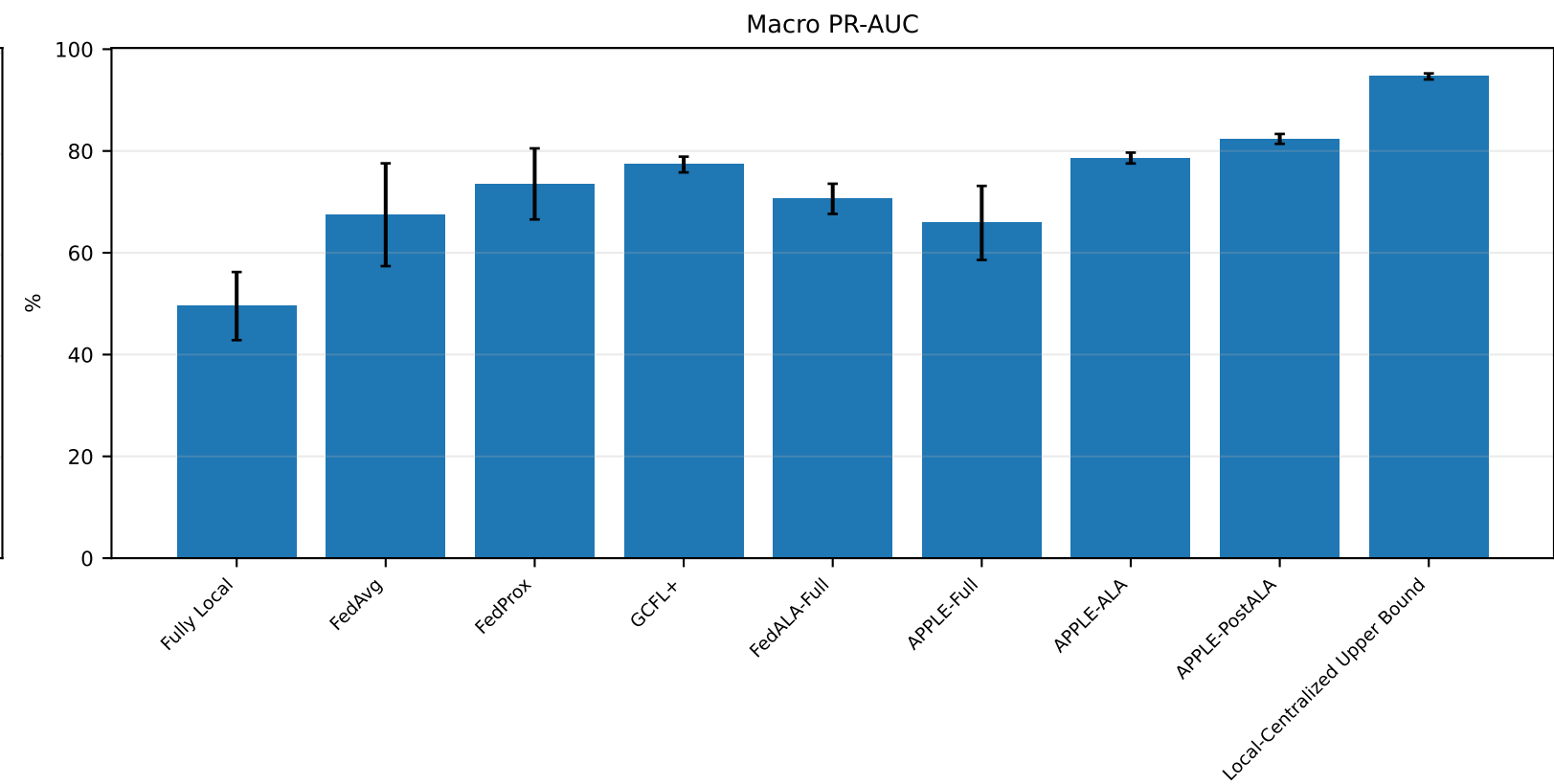
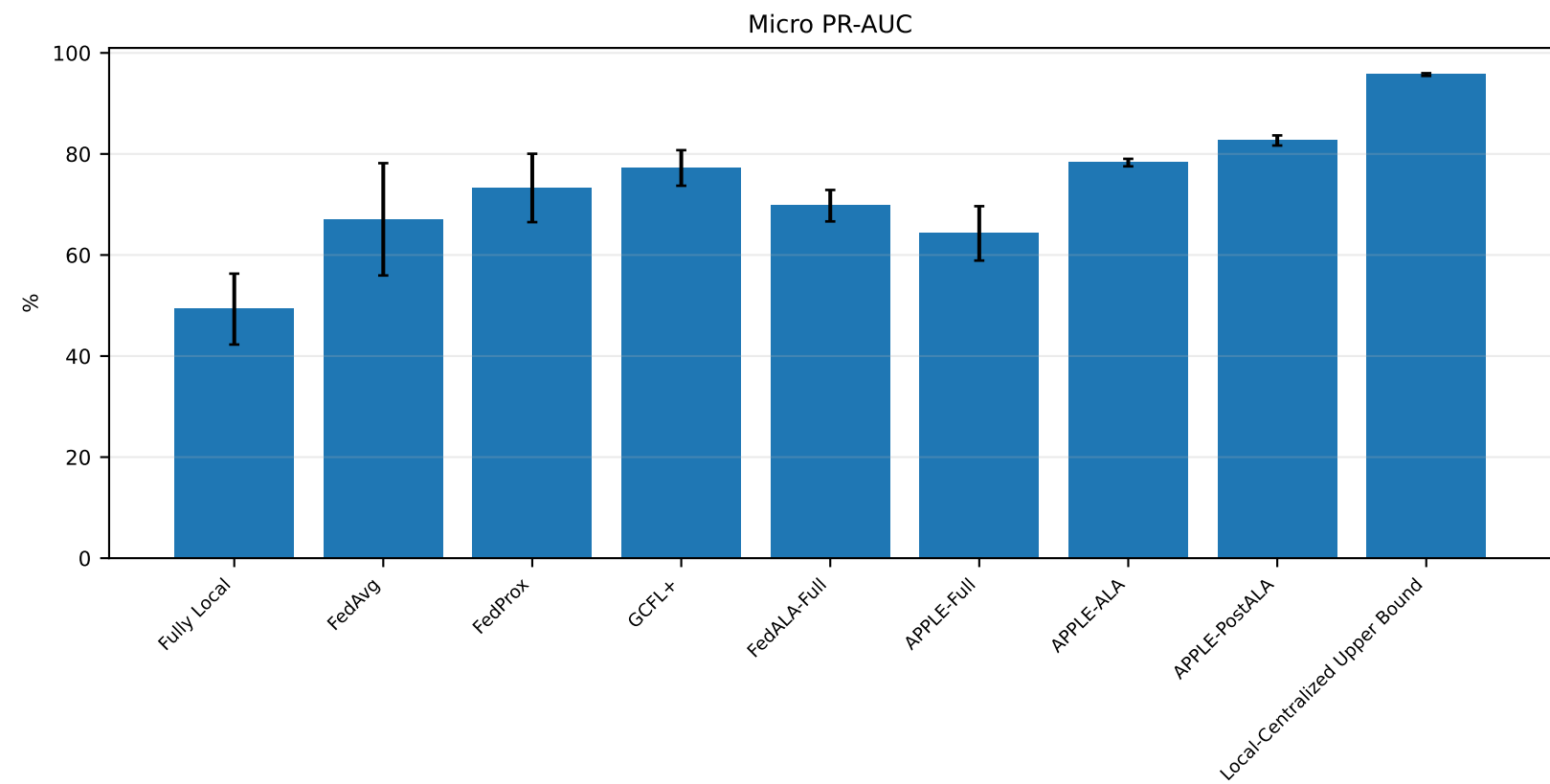
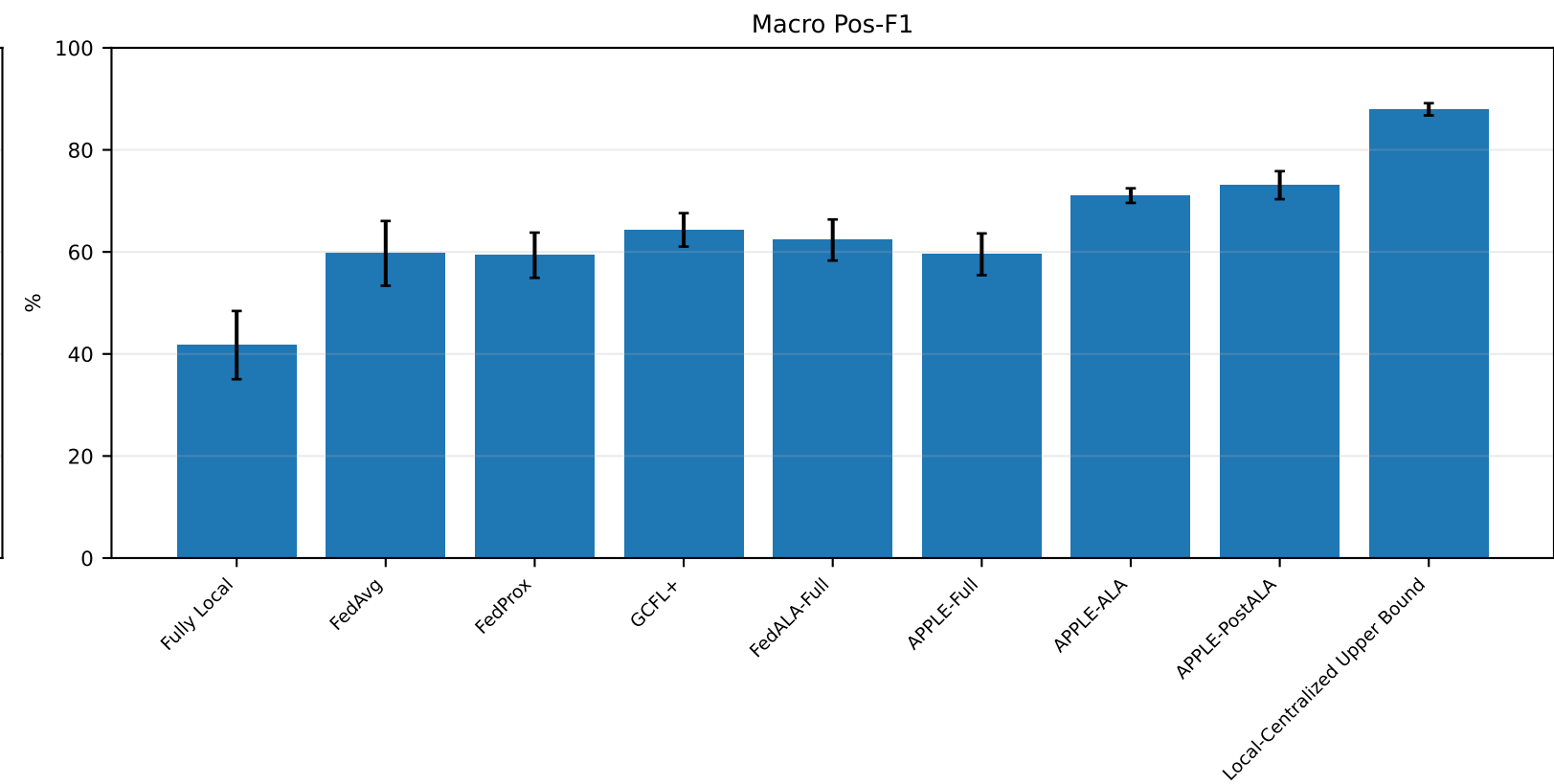
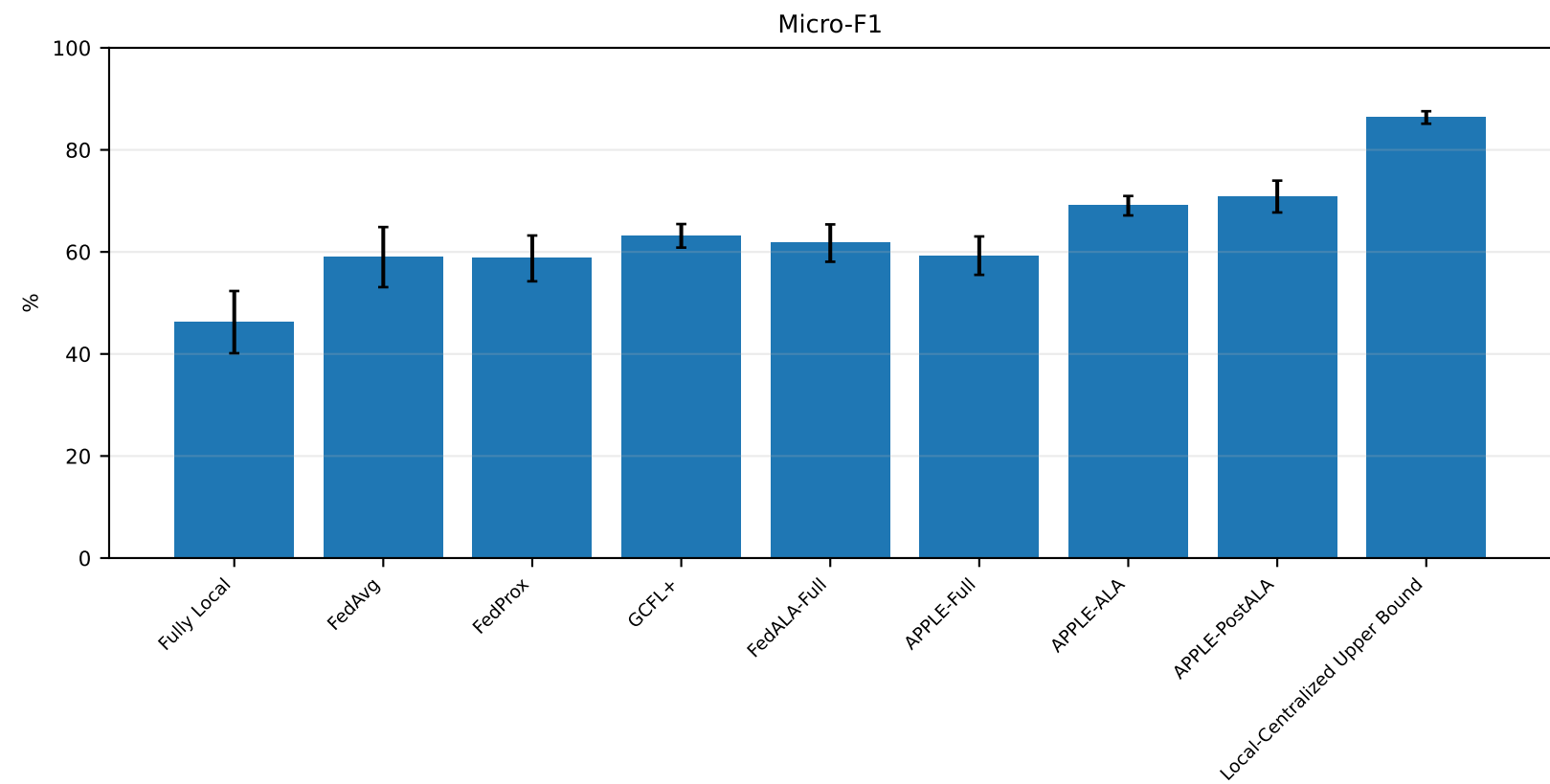
Main baselines + APPLE/FedALA/PostALA | High heterogeneity ($q=0.8$) | selector=macro_pos_f1 | protocol=VO | overview bars
V+O: visible validation selection → oracle/full test



Main baselines + APPLE/FedALA/PostALA | High heterogeneity (q=0.8) | selector=micro_pr_auc | protocol=OO
O+O: oracle/full validation selection → oracle/full test

Method	Micro-F1	Macro Pos-F1	Micro PR-AUC	Macro PR-AUC	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1	Sel. round	n
Fully Local	46.2% ± 6.1%	41.7% ± 6.7%	49.3% ± 7.0%	49.5% ± 6.7%	42.5% ± 10.9%	44.9% ± 8.8%	43.0% ± 7.0%	40.9% ± 7.1%	37.4% ± 0.3%	—	3
FedAvg	59.0% ± 5.9%	59.7% ± 6.3%	67.1% ± 11.1%	67.5% ± 10.1%	60.7% ± 8.4%	64.6% ± 8.9%	60.9% ± 4.9%	60.6% ± 7.7%	51.8% ± 3.1%	72.0 ± 6.2	3
FedProx	58.7% ± 4.5%	59.4% ± 4.4%	73.3% ± 6.8%	73.5% ± 7.0%	59.1% ± 5.0%	65.0% ± 3.3%	61.5% ± 2.6%	59.3% ± 7.3%	51.8% ± 4.4%	74.0 ± 5.3	3
GCFL+	63.2% ± 2.3%	64.3% ± 3.3%	77.2% ± 3.5%	77.3% ± 1.6%	67.2% ± 13.6%	70.6% ± 2.9%	66.5% ± 0.6%	63.2% ± 1.6%	54.1% ± 1.0%	77.7 ± 2.5	3
FedALA-Full	61.7% ± 3.7%	62.3% ± 4.0%	69.8% ± 3.1%	70.6% ± 3.0%	63.6% ± 8.2%	68.3% ± 6.8%	63.6% ± 1.4%	62.1% ± 3.7%	54.1% ± 2.7%	79.3 ± 1.2	3
APPLE-Full	59.3% ± 3.8%	59.5% ± 4.1%	64.3% ± 5.4%	65.9% ± 7.3%	61.4% ± 9.1%	61.6% ± 5.2%	58.3% ± 3.8%	62.2% ± 5.3%	54.2% ± 1.0%	78.0 ± 2.0	3
APPLE-ALA	69.1% ± 1.9%	71.0% ± 1.4%	78.3% ± 0.7%	78.6% ± 1.1%	82.2% ± 2.9%	79.4% ± 3.3%	68.8% ± 2.1%	68.2% ± 4.9%	56.5% ± 3.4%	75.3 ± 2.9	3
APPLE-PostALA	70.9% ± 3.1%	73.1% ± 2.7%	82.7% ± 1.0%	82.4% ± 1.0%	84.4% ± 0.8%	83.8% ± 3.9%	69.9% ± 0.6%	69.4% ± 5.2%	57.9% ± 4.3%	77.3 ± 2.9	3
Local-Centralized Upper Bound	86.3% ± 1.2%	87.9% ± 1.2%	95.7% ± 0.3%	94.6% ± 0.6%	97.5% ± 1.3%	95.5% ± 1.1%	88.7% ± 3.9%	83.1% ± 1.2%	75.0% ± 2.1%	—	3

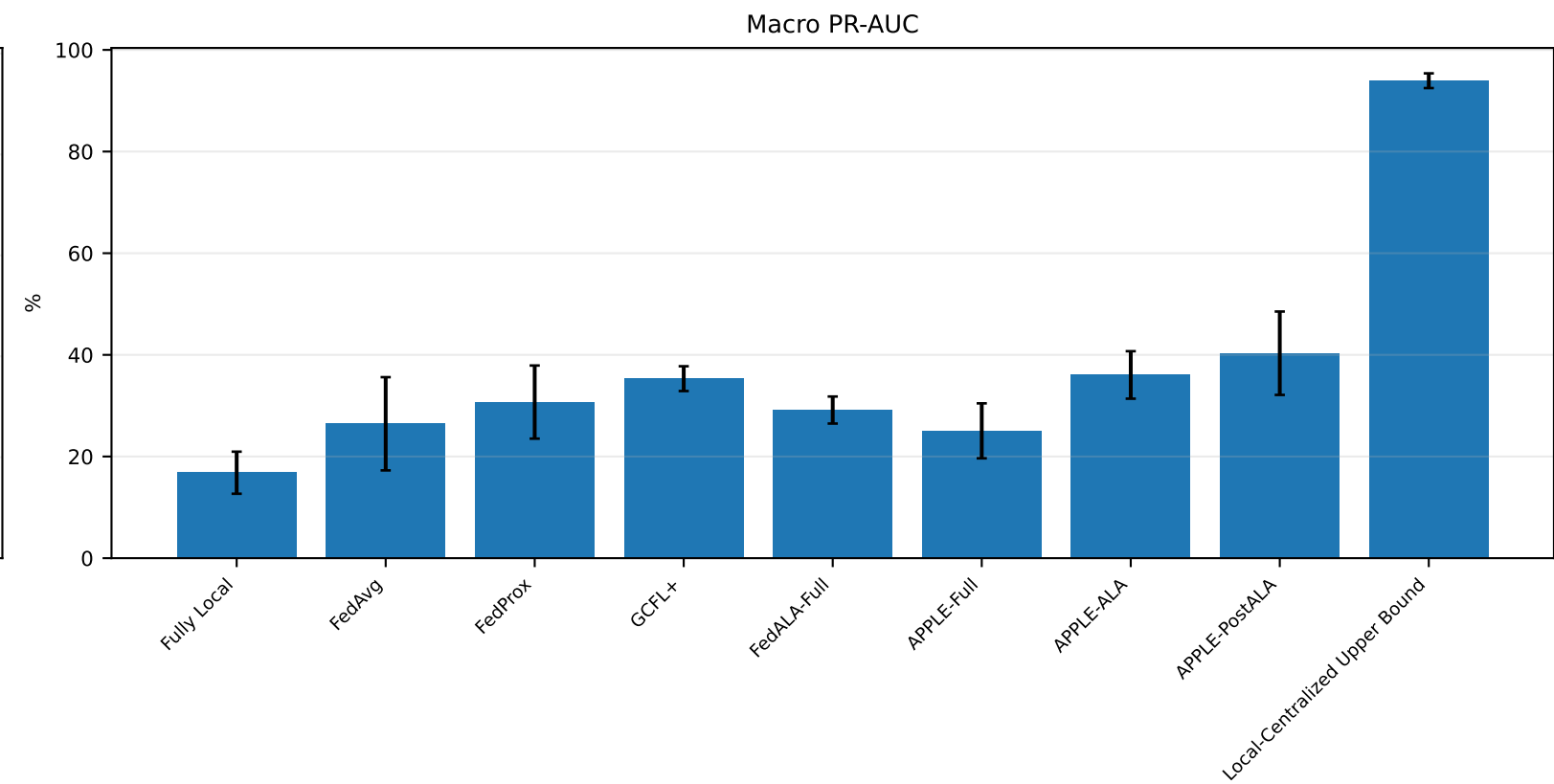
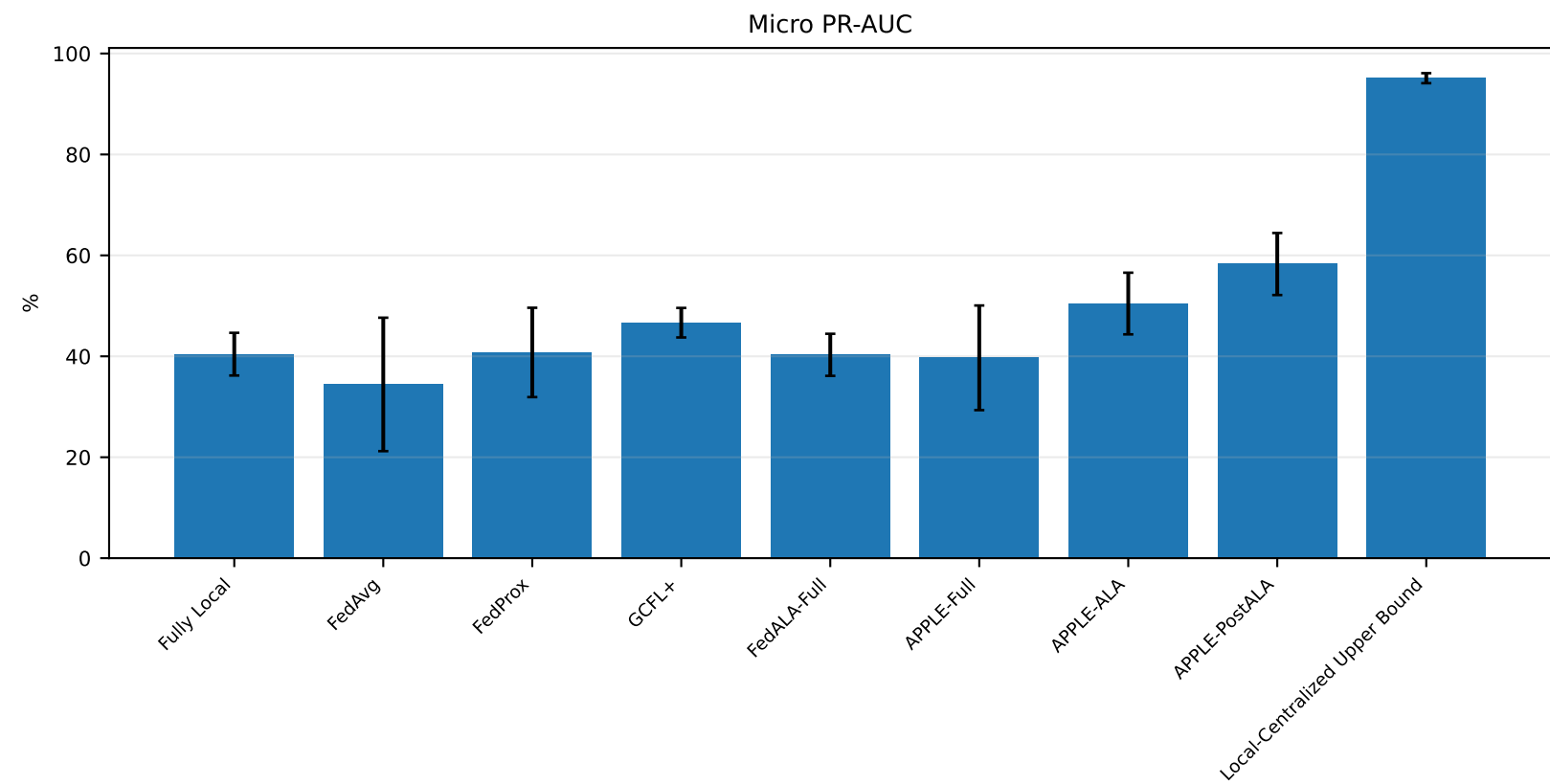
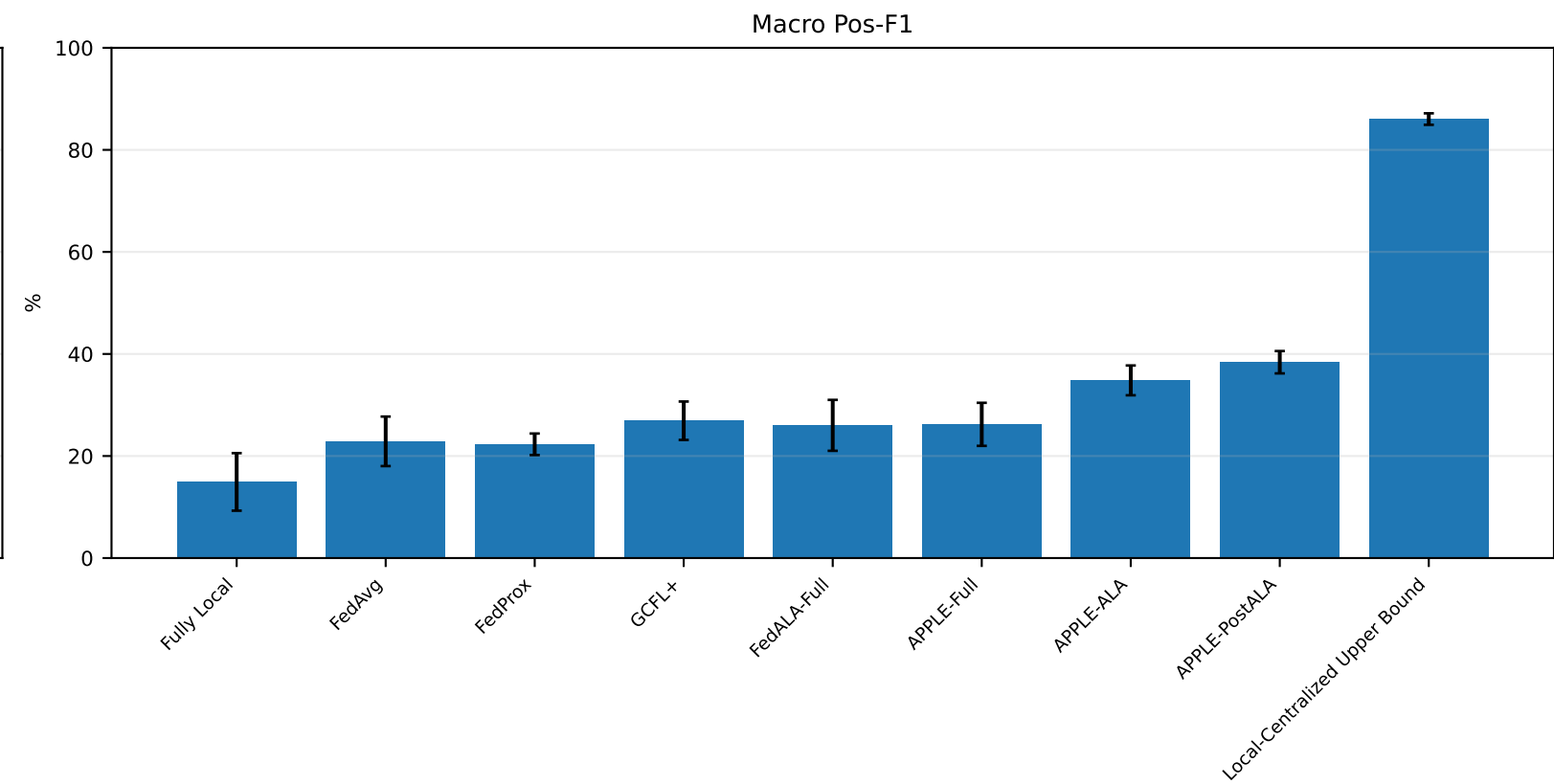
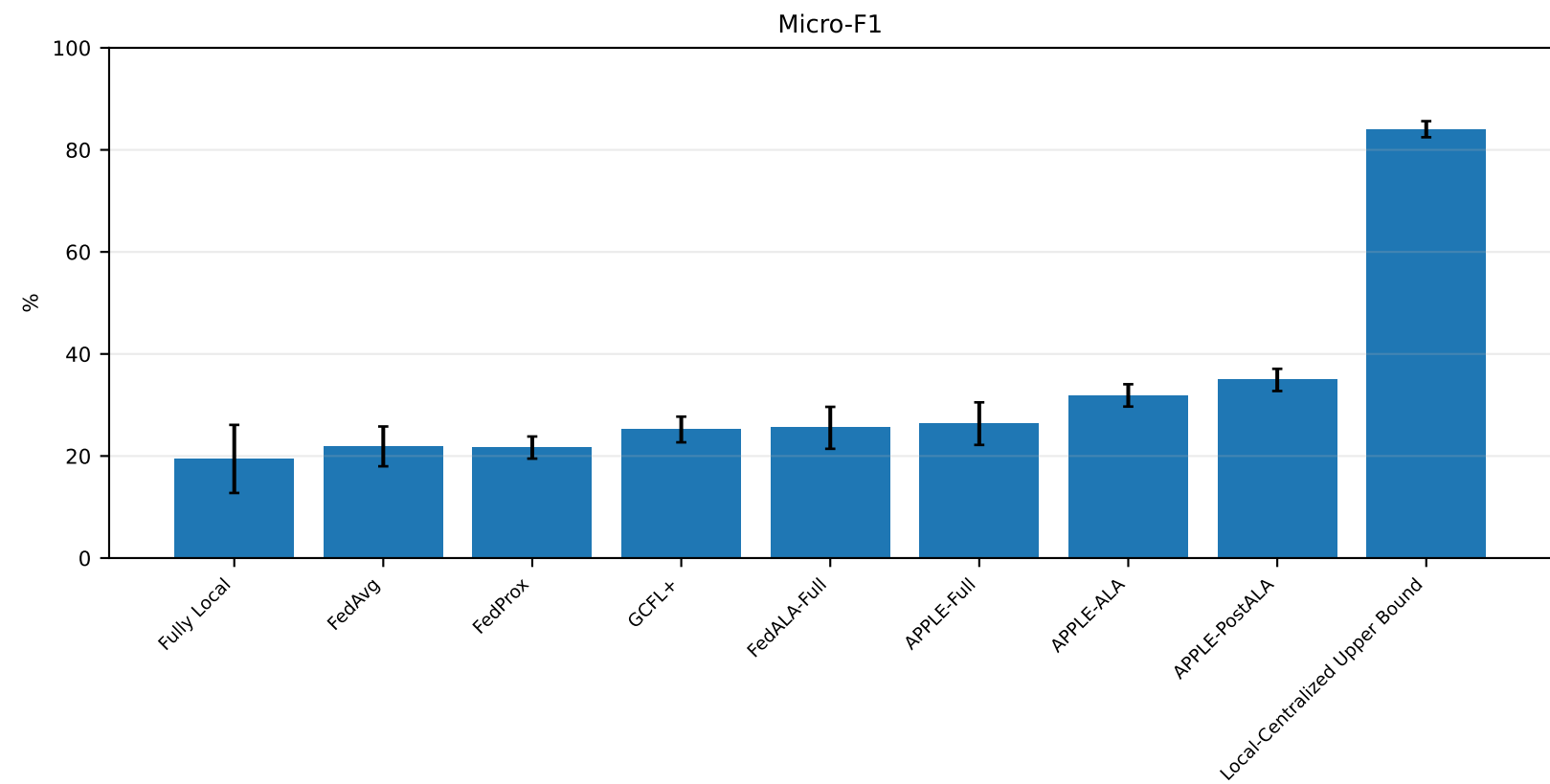
Main baselines + APPLE/FedALA/PostALA | High heterogeneity ($q=0.8$) | selector=micro_pr_auc | protocol=OO | overview bars
O+O: oracle/full validation selection → oracle/full test



Main baselines + APPLE/FedALA/PostALA | High heterogeneity (q=0.8) | selector=micro_pr_auc | protocol=VV
V+V: visible validation selection → visible test

Method	Micro-F1	Macro Pos-F1	Micro PR-AUC	Macro PR-AUC	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1	Sel. round	n
Fully Local	19.4% ± 6.7%	14.9% ± 5.6%	40.4% ± 4.2%	16.8% ± 4.1%	12.6% ± 3.9%	13.9% ± 6.0%	13.6% ± 4.1%	20.3% ± 15.3%	14.2% ± 2.7%	—	3
FedAvg	21.9% ± 3.9%	22.9% ± 4.9%	34.4% ± 13.2%	26.4% ± 9.2%	23.0% ± 7.6%	27.5% ± 9.2%	24.3% ± 2.6%	22.5% ± 4.1%	17.1% ± 1.4%	71.7 ± 11.0	3
FedProx	21.7% ± 2.2%	22.3% ± 2.1%	40.8% ± 8.9%	30.7% ± 7.2%	22.2% ± 2.3%	26.4% ± 1.5%	23.0% ± 1.0%	22.2% ± 4.5%	17.6% ± 1.9%	74.0 ± 5.6	3
GCFL+	25.2% ± 2.5%	26.9% ± 3.8%	46.7% ± 2.9%	35.3% ± 2.4%	30.2% ± 14.8%	32.8% ± 2.9%	26.5% ± 3.0%	25.8% ± 3.1%	19.3% ± 0.9%	72.7 ± 7.5	3
FedALA-Full	25.5% ± 4.1%	26.0% ± 5.0%	40.3% ± 4.2%	29.1% ± 2.7%	27.3% ± 6.9%	31.2% ± 7.5%	25.6% ± 3.2%	25.2% ± 4.8%	20.7% ± 3.3%	78.7 ± 2.3	3
APPLE-Full	26.3% ± 4.2%	26.2% ± 4.2%	39.7% ± 10.4%	25.1% ± 5.4%	25.5% ± 6.2%	25.9% ± 3.7%	24.1% ± 3.9%	29.3% ± 4.7%	26.2% ± 3.5%	77.7 ± 3.2	3
APPLE-ALA	31.9% ± 2.2%	34.8% ± 2.9%	50.5% ± 6.1%	36.0% ± 4.7%	44.6% ± 5.4%	42.4% ± 13.2%	34.3% ± 3.9%	30.6% ± 1.0%	22.3% ± 2.7%	73.7 ± 6.7	3
APPLE-PostALA	34.9% ± 2.2%	38.4% ± 2.2%	58.3% ± 6.1%	40.3% ± 8.2%	49.9% ± 7.1%	49.2% ± 4.8%	34.2% ± 5.5%	34.3% ± 3.2%	24.3% ± 2.9%	70.0 ± 6.6	3
Local-Centralized Upper Bound	84.1% ± 1.6%	86.0% ± 1.1%	95.1% ± 1.0%	93.9% ± 1.4%	98.6% ± 0.5%	94.4% ± 1.1%	85.6% ± 3.7%	78.6% ± 2.4%	73.0% ± 3.6%	—	3

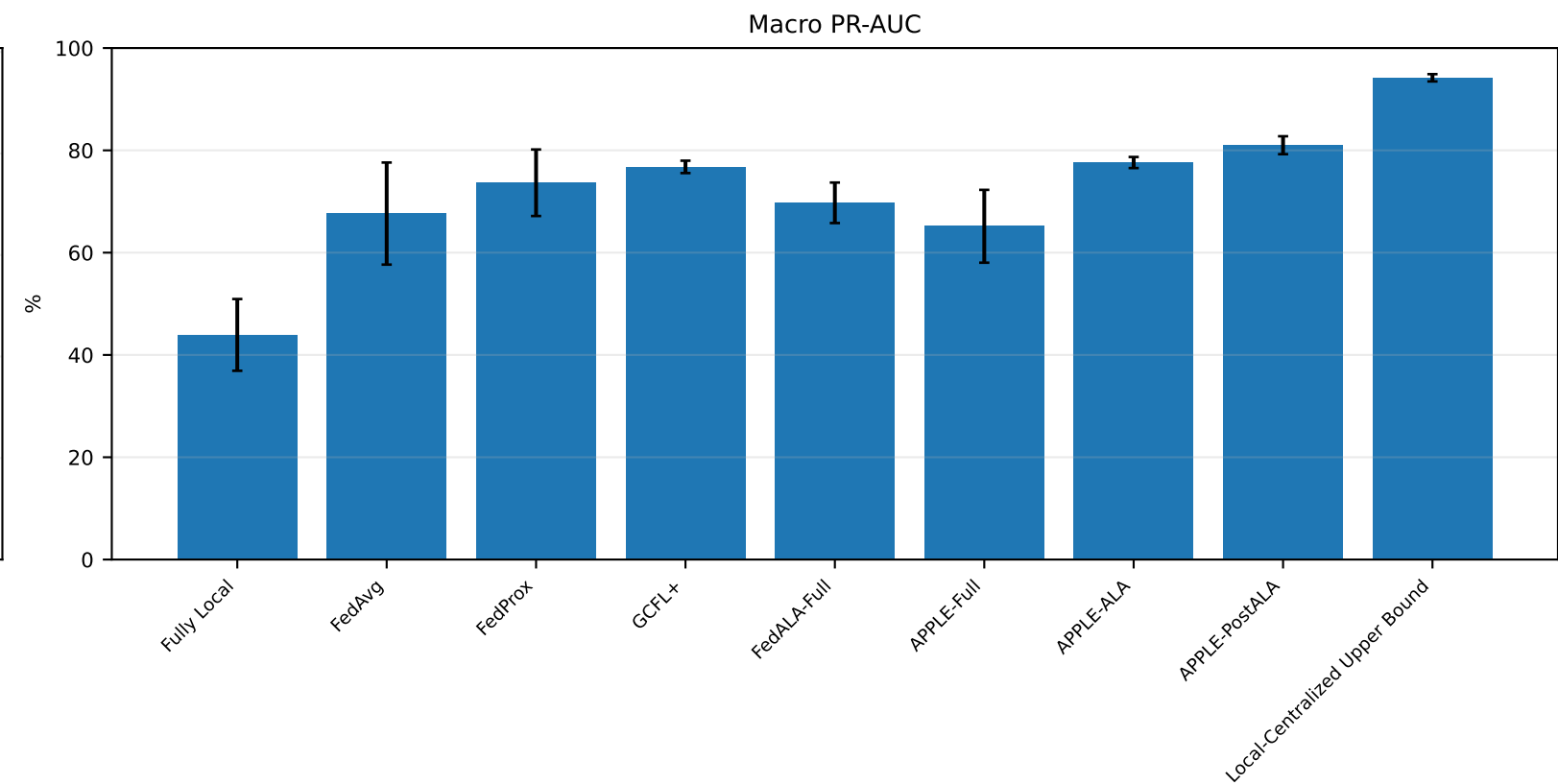
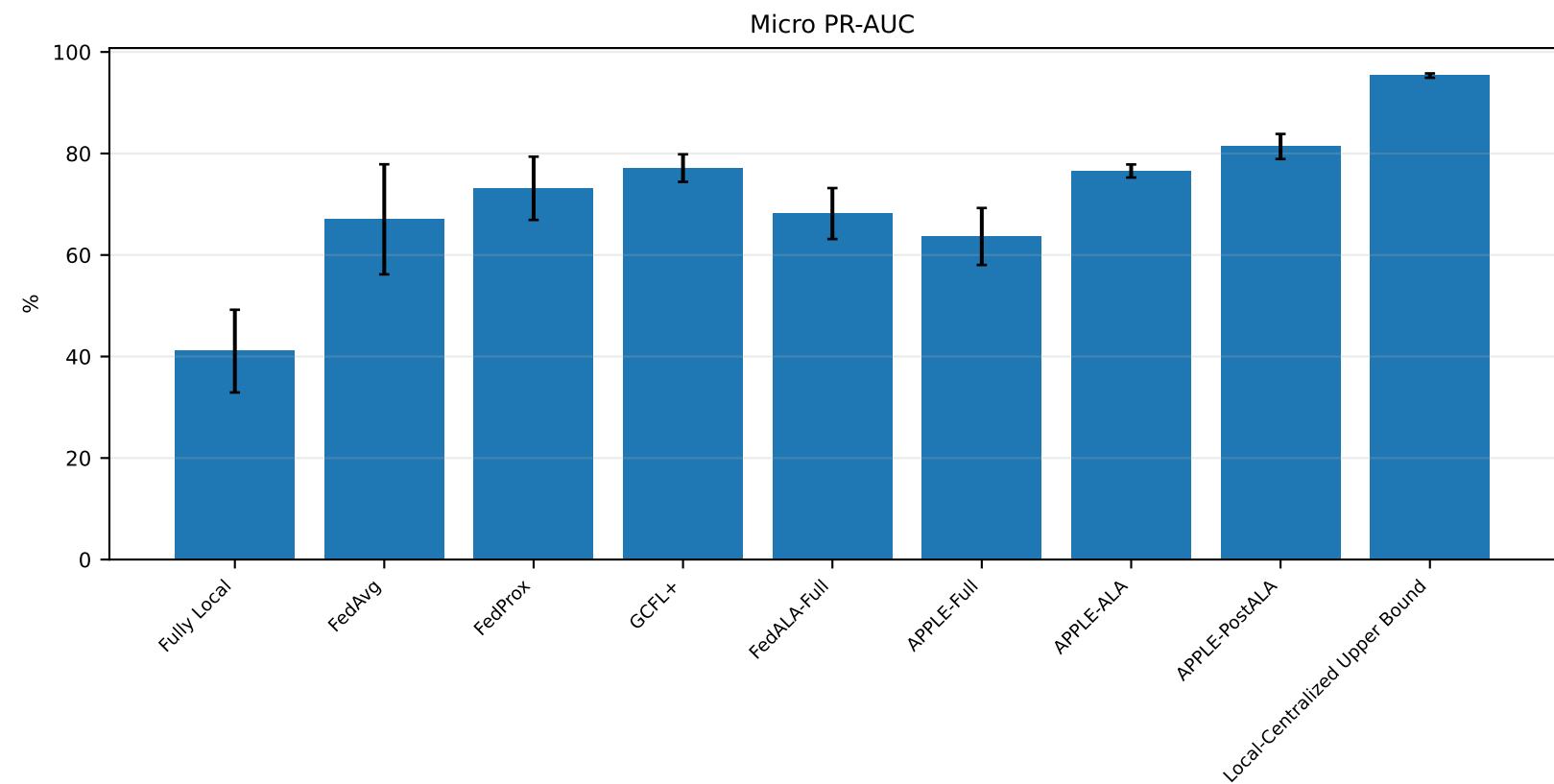
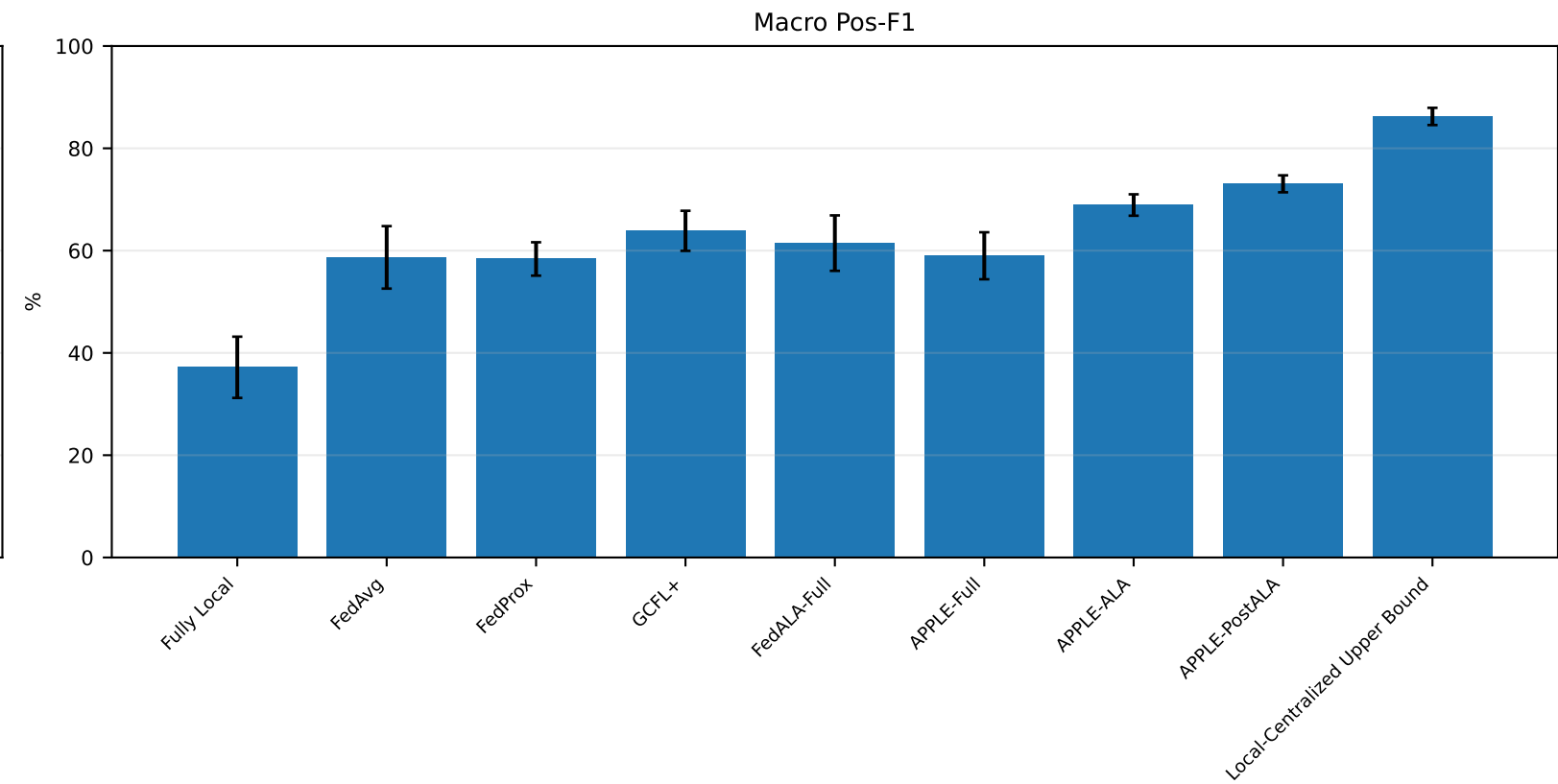
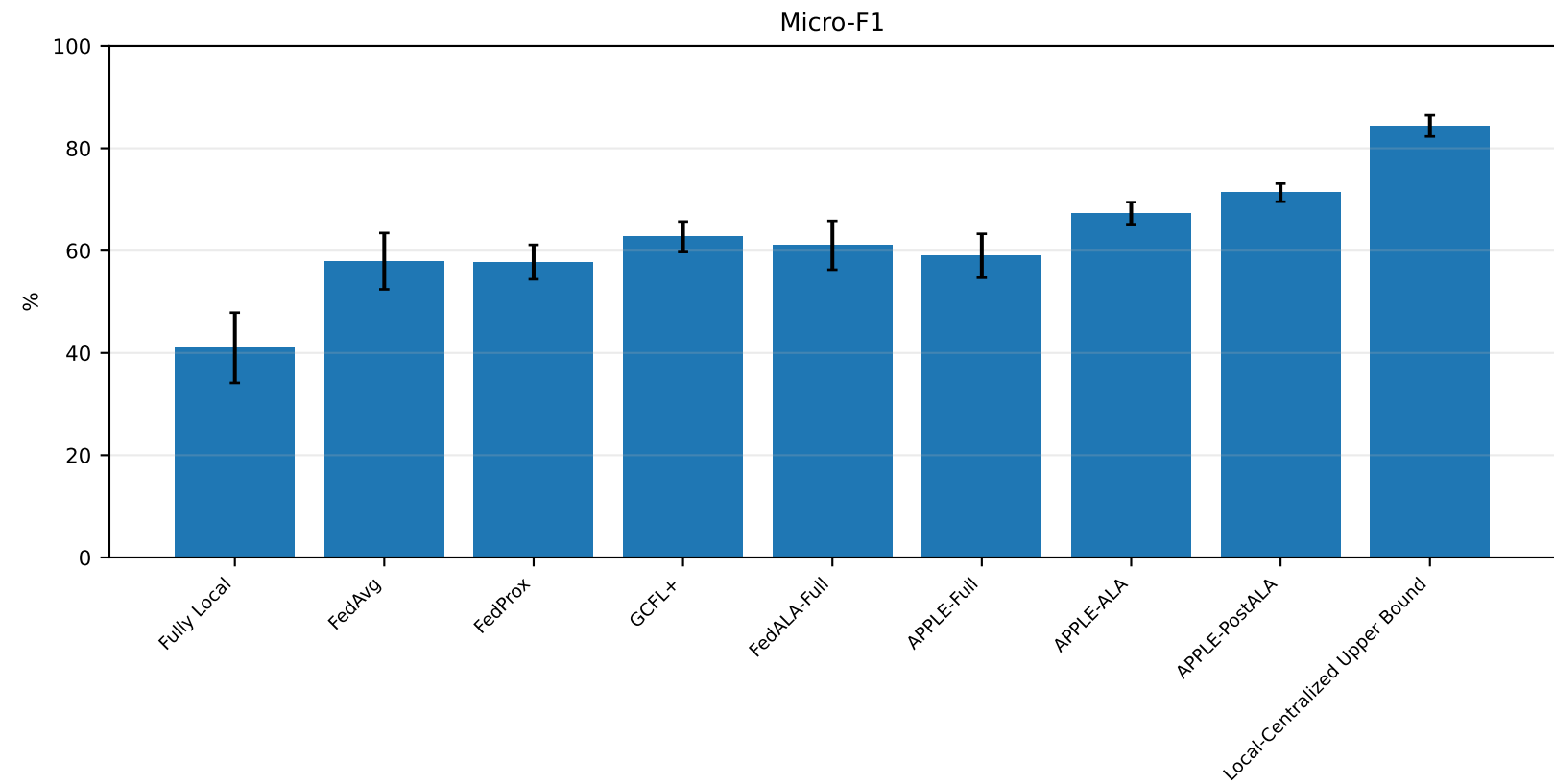
Main baselines + APPLE/FedALA/PostALA | High heterogeneity ($q=0.8$) | selector=micro_pr_auc | protocol=VV | overview bars
V+V: visible validation selection → visible test



Main baselines + APPLE/FedALA/PostALA | High heterogeneity (q=0.8) | selector=micro_pr_auc | protocol=VO
V+O: visible validation selection → oracle/full test

Method	Micro-F1	Macro Pos-F1	Micro PR-AUC	Macro PR-AUC	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1	Sel. round	n
Fully Local	41.0% ± 6.9%	37.2% ± 6.0%	41.1% ± 8.2%	43.9% ± 7.0%	36.4% ± 7.8%	39.2% ± 8.9%	38.9% ± 7.4%	37.8% ± 8.8%	33.6% ± 3.0%	—	3
FedAvg	57.9% ± 5.5%	58.7% ± 6.1%	67.0% ± 10.8%	67.6% ± 10.0%	58.4% ± 9.9%	64.3% ± 10.2%	61.3% ± 3.4%	58.7% ± 5.5%	50.7% ± 2.6%	71.7 ± 11.0	3
FedProx	57.8% ± 3.4%	58.4% ± 3.3%	73.1% ± 6.2%	73.7% ± 6.5%	58.4% ± 3.2%	64.0% ± 2.4%	59.9% ± 1.7%	58.3% ± 6.6%	51.3% ± 3.5%	74.0 ± 5.6	3
GCFL+	62.7% ± 3.0%	63.9% ± 3.9%	77.1% ± 2.7%	76.8% ± 1.2%	66.5% ± 14.6%	71.2% ± 3.9%	64.1% ± 3.2%	63.5% ± 3.2%	54.1% ± 1.0%	72.7 ± 7.5	3
FedALA-Full	61.0% ± 4.8%	61.5% ± 5.4%	68.2% ± 5.0%	69.7% ± 4.0%	63.6% ± 8.3%	67.4% ± 7.6%	61.6% ± 3.1%	60.8% ± 5.0%	54.0% ± 3.3%	78.7 ± 2.3	3
APPLE-Full	59.0% ± 4.3%	59.0% ± 4.6%	63.6% ± 5.6%	65.2% ± 7.1%	58.1% ± 9.7%	58.8% ± 3.5%	58.7% ± 4.1%	63.7% ± 5.4%	55.7% ± 1.9%	77.7 ± 3.2	3
APPLE-ALA	67.3% ± 2.2%	68.9% ± 2.1%	76.6% ± 1.3%	77.6% ± 1.1%	77.3% ± 4.4%	74.8% ± 5.9%	69.7% ± 3.1%	67.0% ± 1.3%	55.7% ± 4.2%	73.7 ± 6.7	3
APPLE-PostALA	71.3% ± 1.8%	73.1% ± 1.7%	81.4% ± 2.5%	81.0% ± 1.8%	82.7% ± 3.7%	81.5% ± 1.4%	72.3% ± 4.7%	70.0% ± 2.1%	58.8% ± 3.8%	70.0 ± 6.6	3
Local-Centralized Upper Bound	84.4% ± 2.1%	86.2% ± 1.7%	95.3% ± 0.4%	94.2% ± 0.7%	97.4% ± 0.2%	94.7% ± 0.1%	86.3% ± 4.8%	79.1% ± 2.4%	73.6% ± 3.6%	—	3

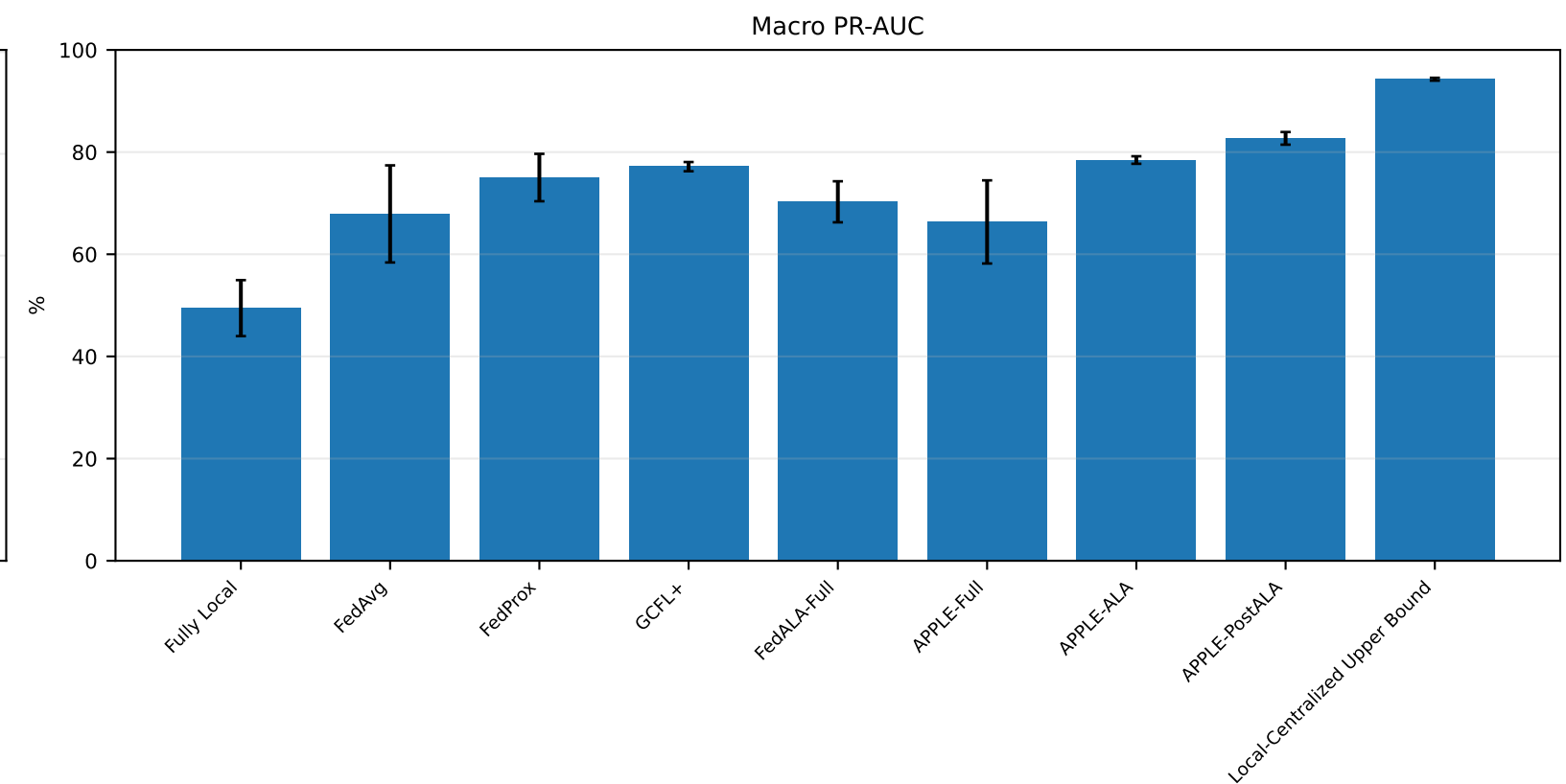
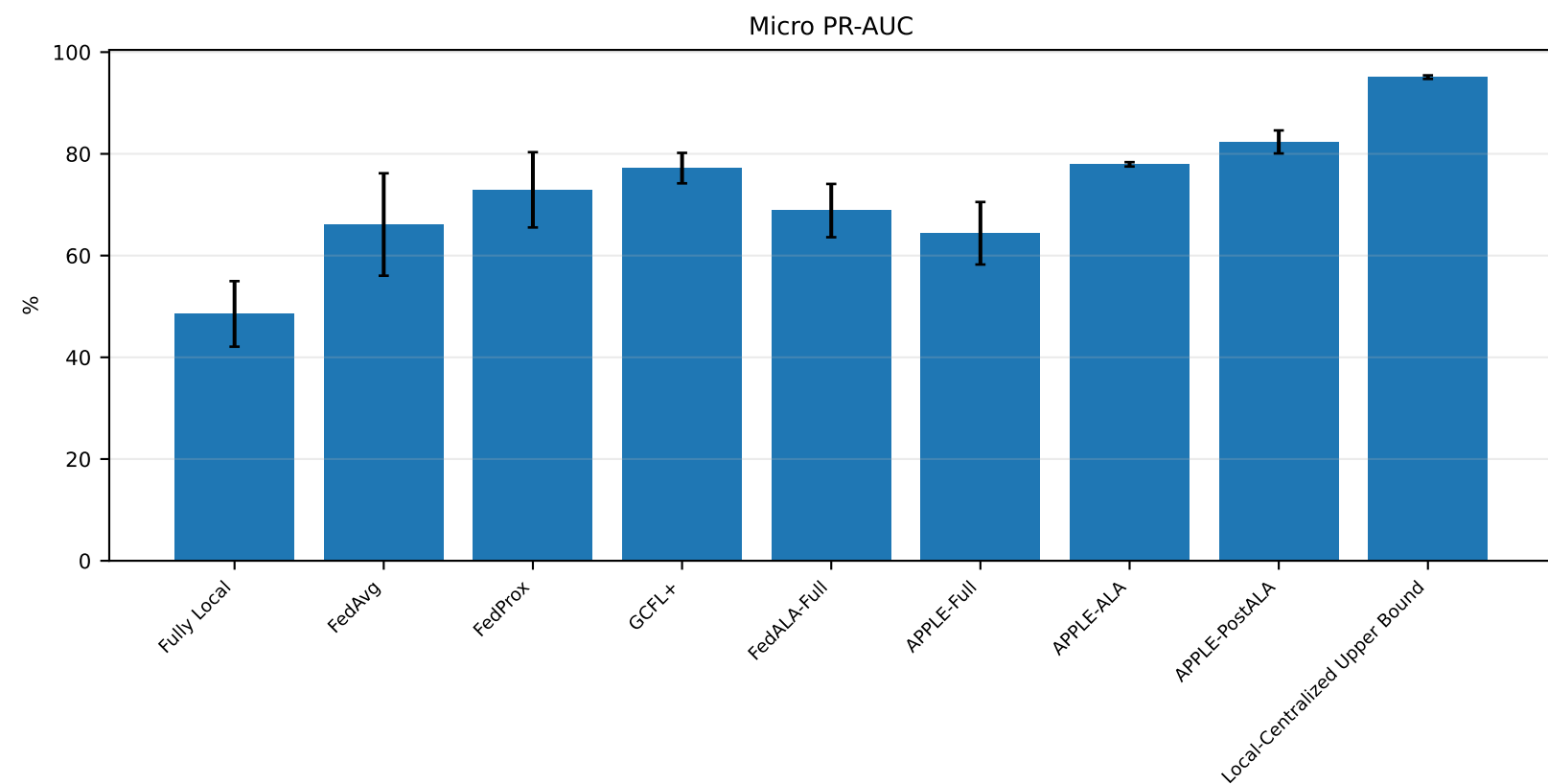
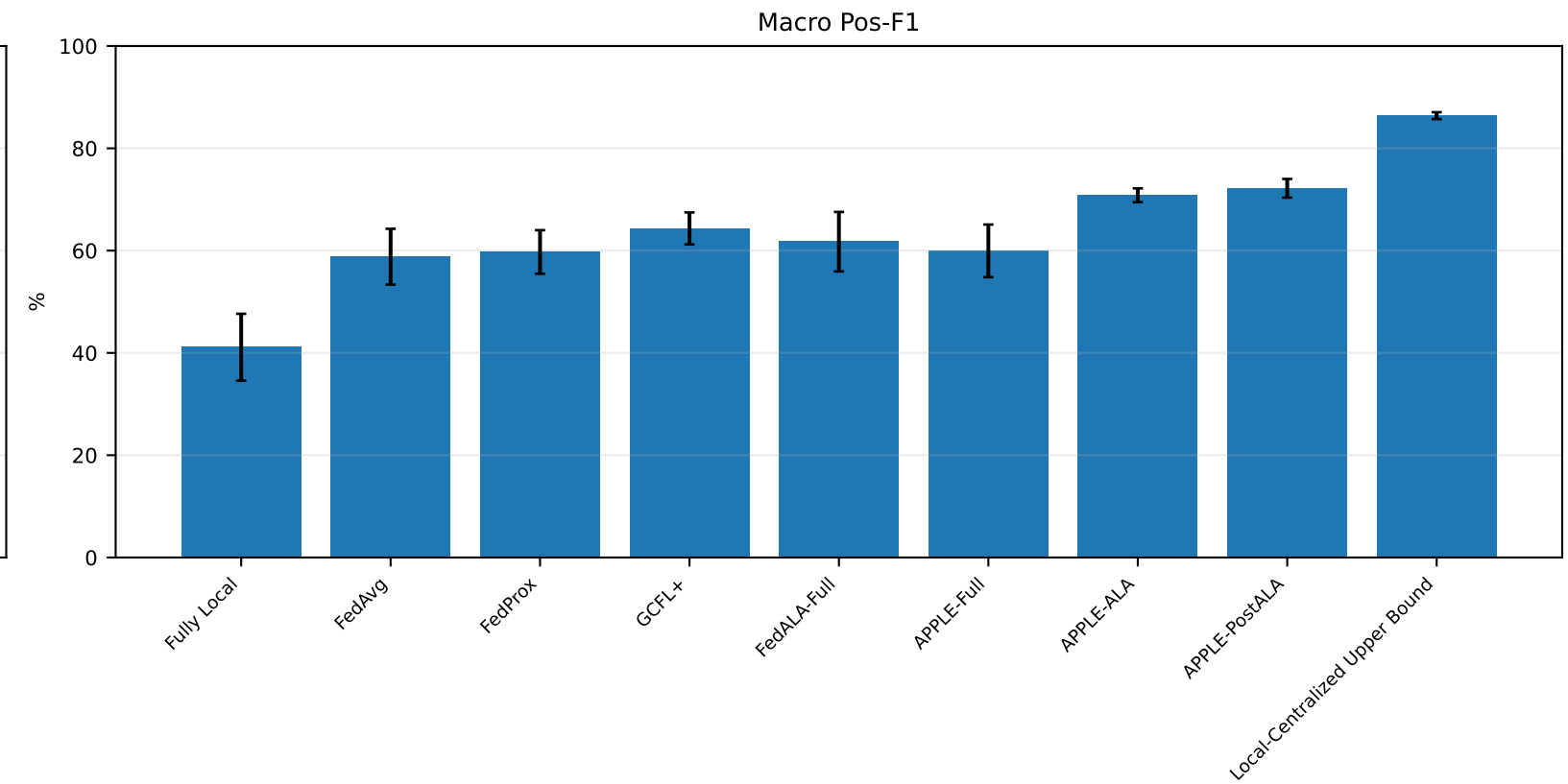
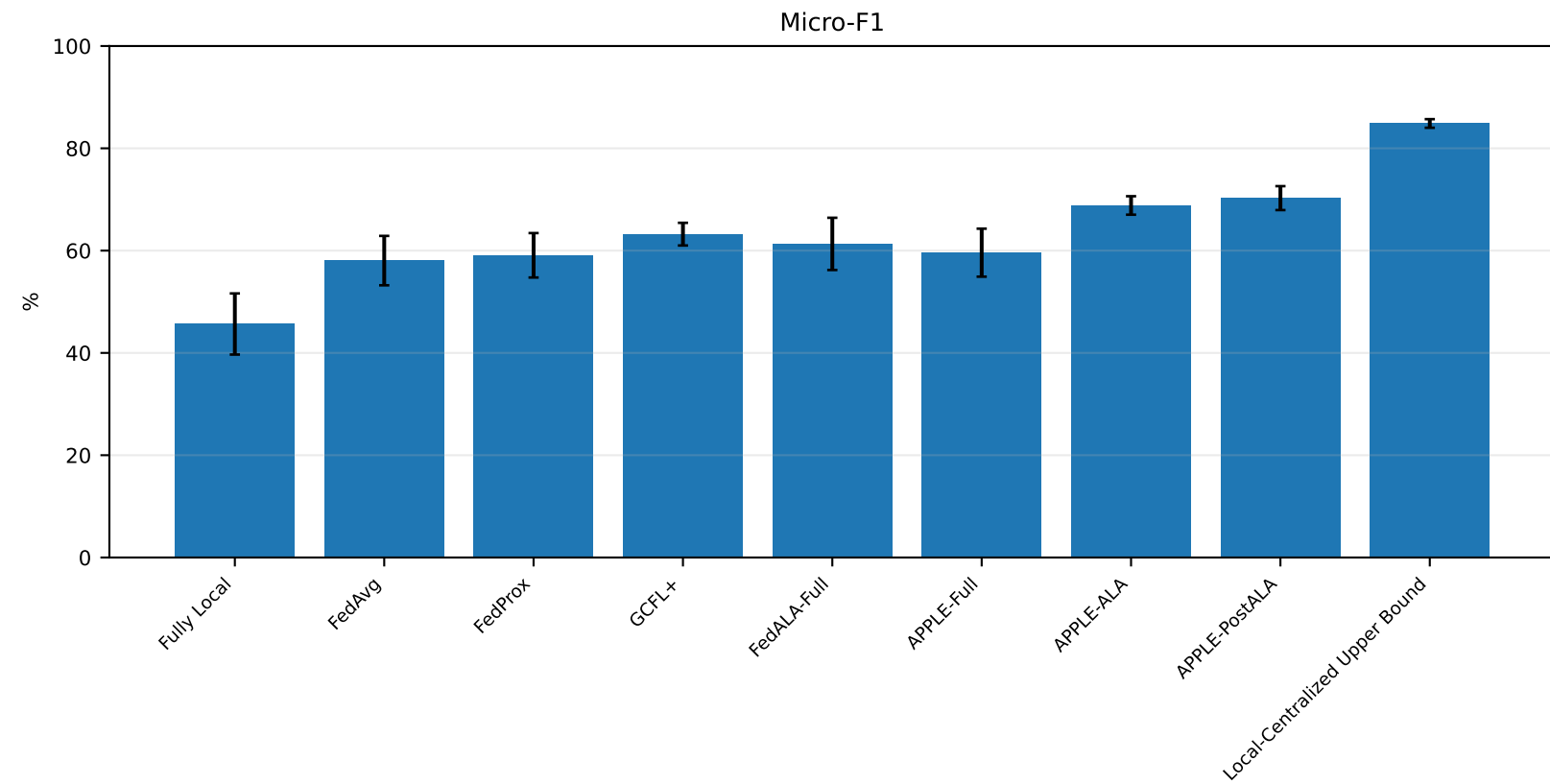
Main baselines + APPLE/FedALA/PostALA | High heterogeneity ($q=0.8$) | selector=micro_pr_auc | protocol=VO | overview bars
V+O: visible validation selection → oracle/full test



Main baselines + APPLE/FedALA/PostALA | High heterogeneity (q=0.8) | selector=macro_pr_auc | protocol=OO
O+O: oracle/full validation selection → oracle/full test

Method	Micro-F1	Macro Pos-F1	Micro PR-AUC	Macro PR-AUC	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1	Sel. round	n
Fully Local	45.7% ± 6.0%	41.1% ± 6.5%	48.5% ± 6.4%	49.5% ± 5.5%	41.6% ± 10.8%	44.3% ± 9.0%	42.3% ± 7.1%	40.5% ± 6.4%	36.8% ± 0.6%	—	3
FedAvg	58.0% ± 4.8%	58.8% ± 5.5%	66.1% ± 10.1%	67.9% ± 9.5%	60.9% ± 10.2%	62.5% ± 6.5%	60.0% ± 4.0%	59.5% ± 6.3%	51.1% ± 2.0%	78.0 ± 2.6	3
FedProx	59.1% ± 4.3%	59.7% ± 4.3%	72.9% ± 7.4%	75.0% ± 4.6%	60.8% ± 3.8%	64.1% ± 4.4%	61.8% ± 2.3%	59.9% ± 6.7%	52.1% ± 4.5%	77.3 ± 1.2	3
GCFL+	63.2% ± 2.2%	64.3% ± 3.1%	77.2% ± 3.0%	77.2% ± 0.9%	67.6% ± 13.1%	70.2% ± 3.6%	66.0% ± 0.6%	63.2% ± 1.8%	54.7% ± 1.7%	79.3 ± 1.2	3
FedALA-Full	61.3% ± 5.1%	61.7% ± 5.8%	68.8% ± 5.2%	70.3% ± 4.0%	64.0% ± 9.1%	67.6% ± 8.1%	61.9% ± 3.4%	61.5% ± 5.6%	53.8% ± 3.2%	78.7 ± 2.3	3
APPLE-Full	59.6% ± 4.7%	60.0% ± 5.1%	64.4% ± 6.1%	66.3% ± 8.1%	62.1% ± 10.9%	63.1% ± 6.8%	57.8% ± 3.0%	62.8% ± 6.8%	53.9% ± 0.9%	79.3 ± 1.2	3
APPLE-ALA	68.8% ± 1.8%	70.8% ± 1.4%	77.9% ± 0.4%	78.4% ± 0.7%	82.0% ± 3.3%	79.4% ± 3.4%	68.2% ± 2.0%	68.0% ± 4.8%	56.5% ± 3.5%	75.3 ± 2.9	3
APPLE-PostALA	70.3% ± 2.3%	72.2% ± 1.8%	82.3% ± 2.3%	82.7% ± 1.2%	82.5% ± 2.9%	81.5% ± 3.4%	70.0% ± 0.9%	69.1% ± 4.7%	57.9% ± 4.3%	76.0 ± 5.2	3
Local-Centralized Upper Bound	84.9% ± 0.8%	86.4% ± 0.7%	95.1% ± 0.4%	94.3% ± 0.3%	96.7% ± 0.2%	93.3% ± 0.5%	86.1% ± 2.2%	81.1% ± 0.2%	74.7% ± 1.5%	—	3

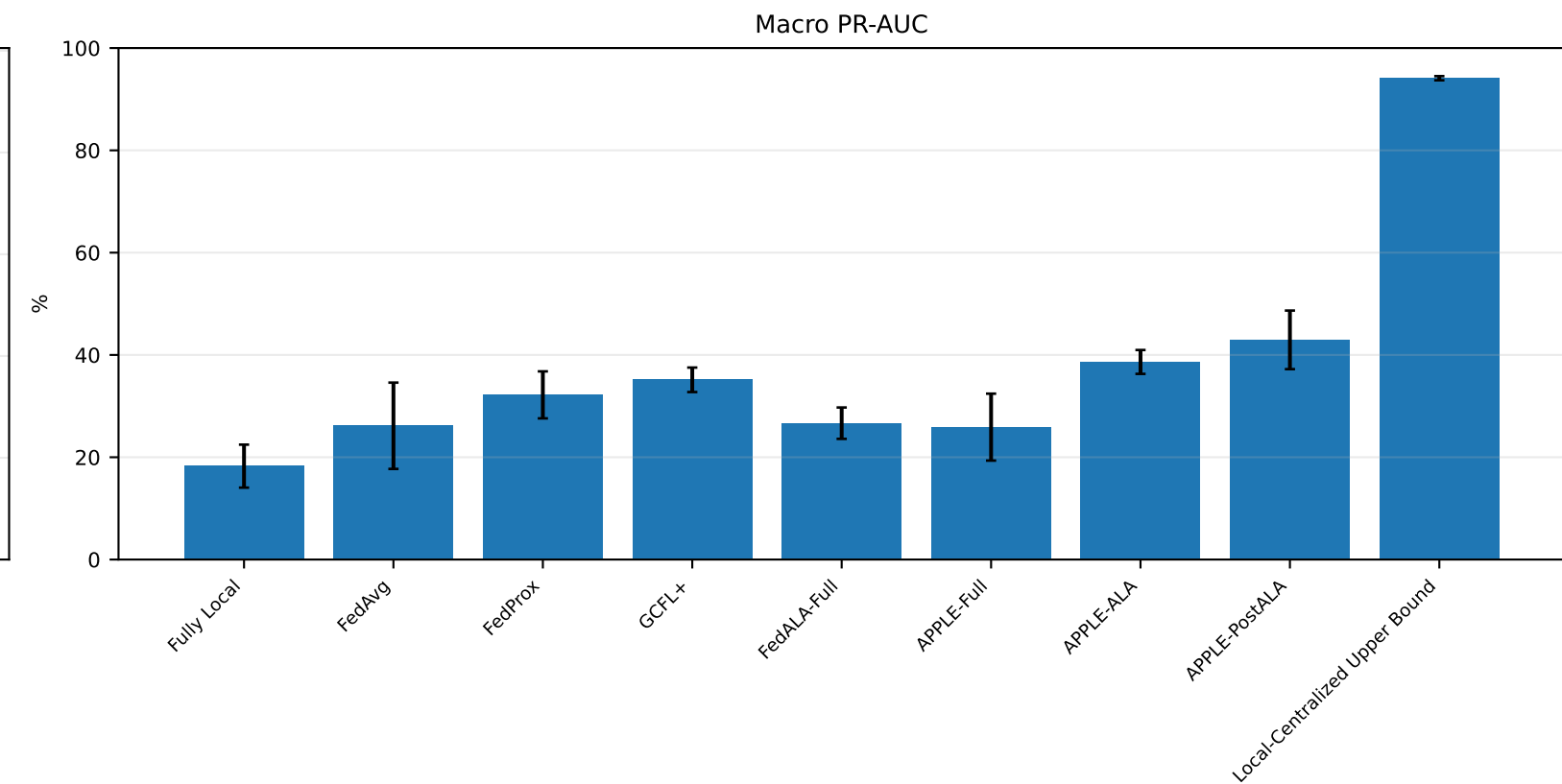
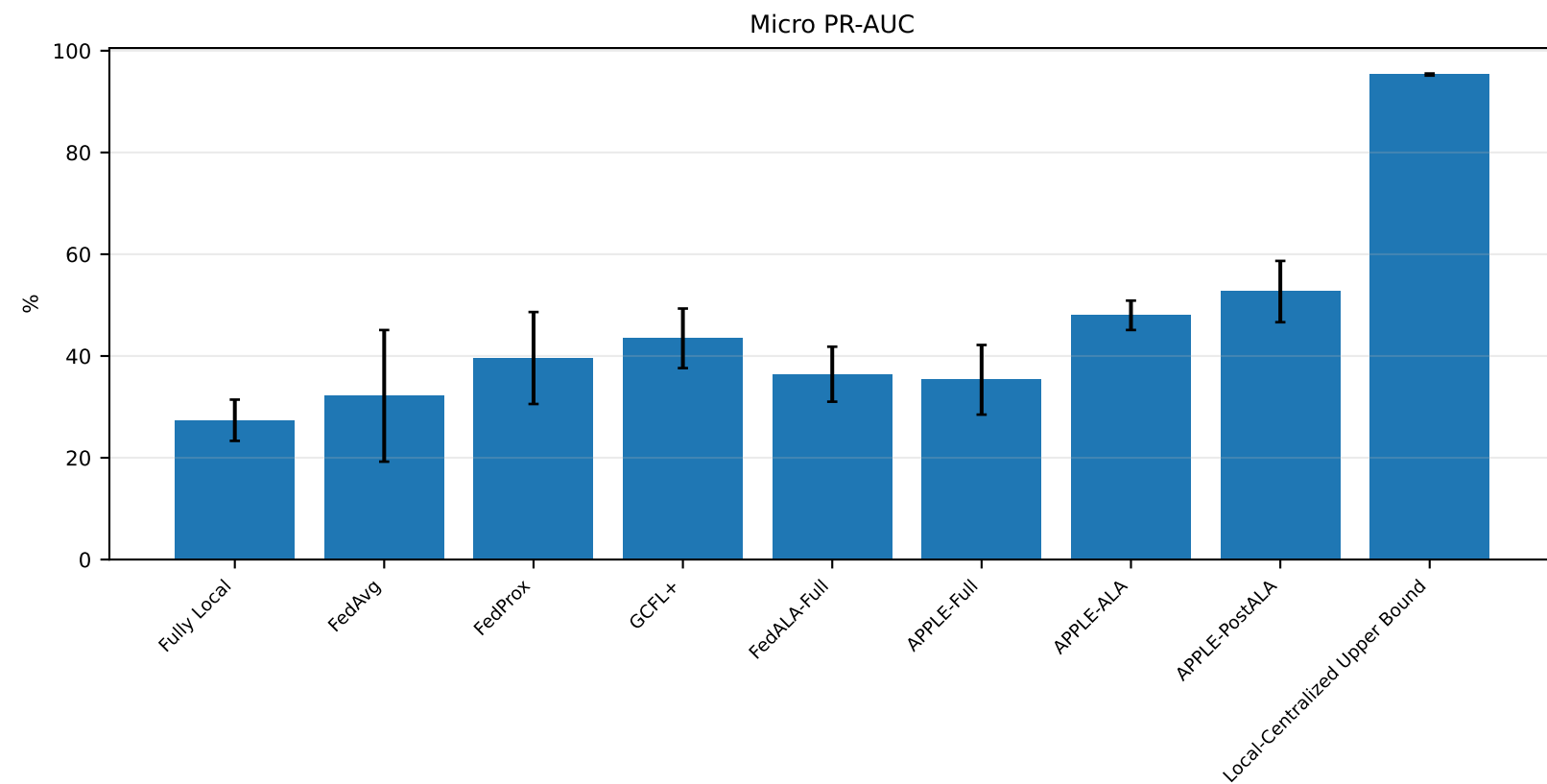
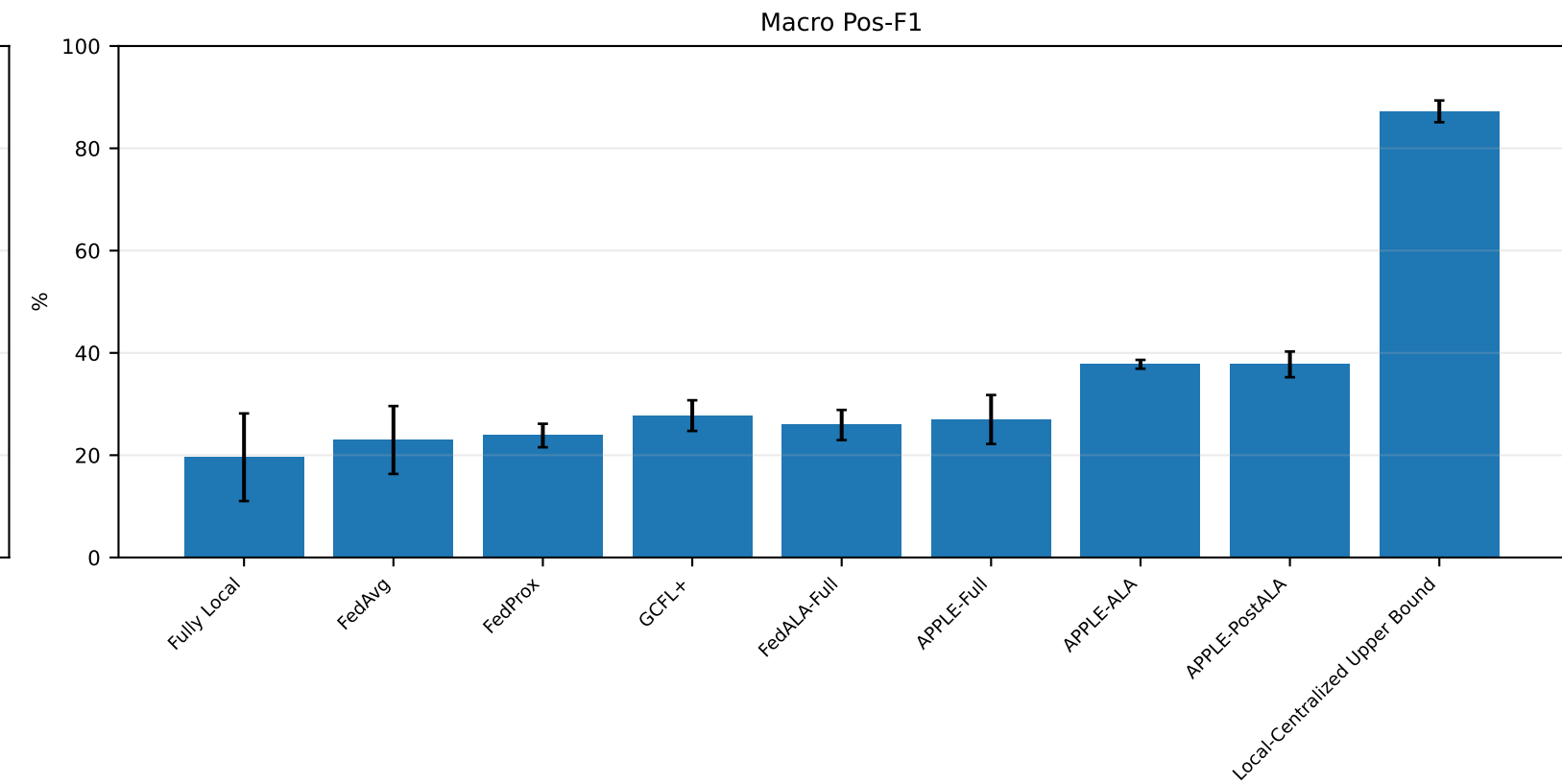
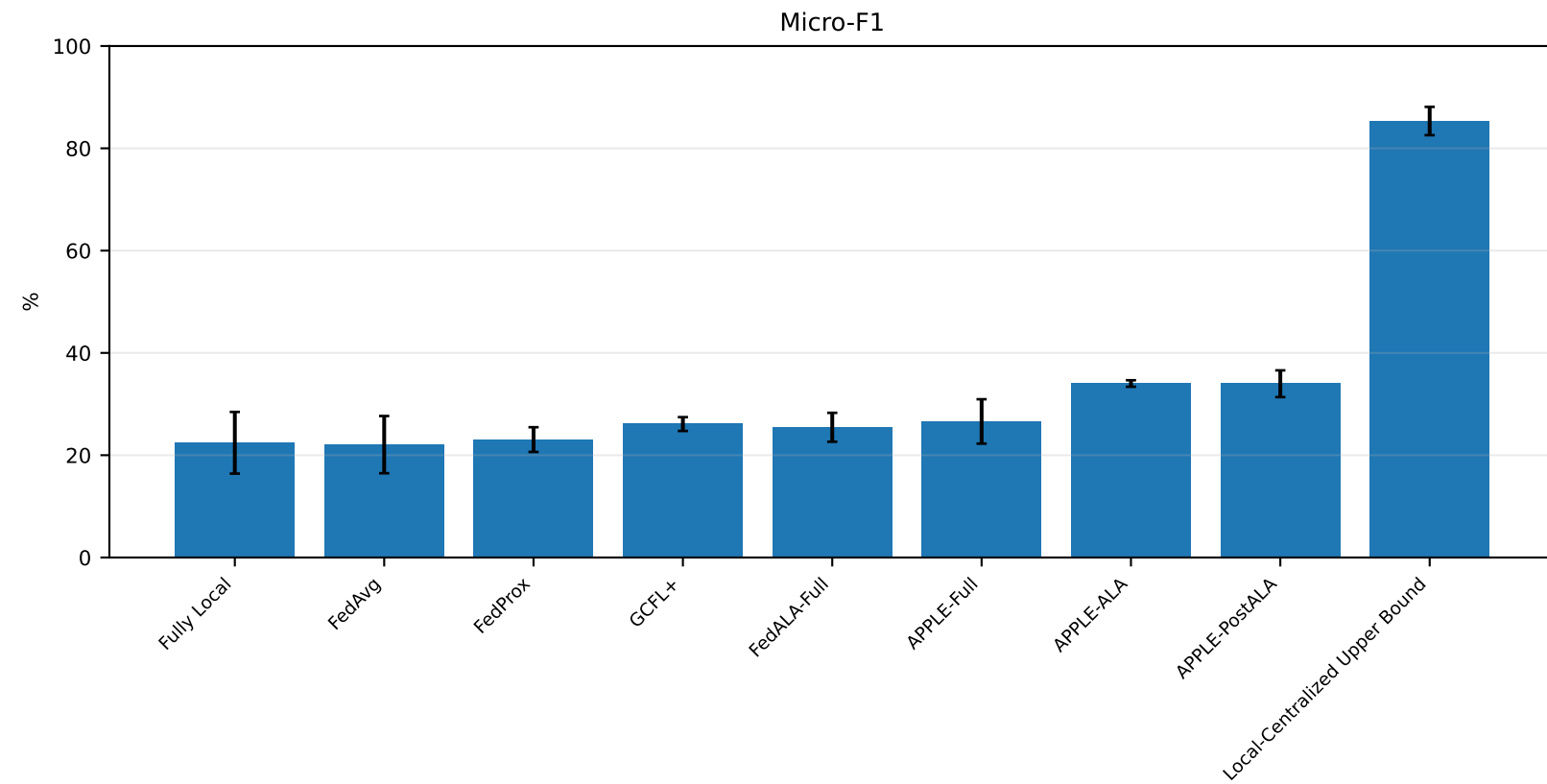
Main baselines + APPLE/FedALA/PostALA | High heterogeneity ($q=0.8$) | selector=macro_pr_auc | protocol=OO | overview bars
O+O: oracle/full validation selection → oracle/full test



Main baselines + APPLE/FedALA/PostALA | High heterogeneity (q=0.8) | selector=macro_pr_auc | protocol=VV
V+V: visible validation selection → visible test

Method	Micro-F1	Macro Pos-F1	Micro PR-AUC	Macro PR-AUC	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1	Sel. round	n
Fully Local	22.4% ± 6.0%	19.6% ± 8.6%	27.4% ± 4.1%	18.3% ± 4.2%	21.8% ± 12.7%	22.0% ± 11.1%	18.3% ± 6.2%	20.3% ± 10.4%	15.7% ± 2.4%	—	3
FedAvg	22.1% ± 5.6%	23.0% ± 6.6%	32.2% ± 12.9%	26.2% ± 8.4%	23.1% ± 11.1%	27.6% ± 10.0%	23.0% ± 3.4%	23.4% ± 6.2%	17.8% ± 2.6%	74.7 ± 2.1	3
FedProx	23.1% ± 2.4%	23.8% ± 2.3%	39.6% ± 9.0%	32.2% ± 4.6%	23.8% ± 3.6%	27.9% ± 4.3%	23.9% ± 2.0%	25.1% ± 4.5%	18.6% ± 2.3%	78.0 ± 1.7	3
GCFL+	26.1% ± 1.4%	27.7% ± 3.0%	43.5% ± 5.9%	35.1% ± 2.4%	32.7% ± 13.3%	30.9% ± 2.8%	28.8% ± 0.8%	26.5% ± 0.6%	19.8% ± 1.1%	76.0 ± 5.3	3
FedALA-Full	25.5% ± 2.8%	25.9% ± 2.9%	36.4% ± 5.4%	26.7% ± 3.1%	26.4% ± 4.7%	29.8% ± 5.6%	26.4% ± 2.5%	26.0% ± 3.3%	20.8% ± 2.5%	78.0 ± 2.6	3
APPLE-Full	26.6% ± 4.3%	27.0% ± 4.8%	35.3% ± 6.8%	25.9% ± 6.5%	29.1% ± 6.7%	29.2% ± 5.7%	23.1% ± 2.1%	29.0% ± 7.3%	24.5% ± 3.3%	79.3 ± 1.2	3
APPLE-ALA	34.0% ± 0.6%	37.8% ± 0.9%	48.0% ± 2.9%	38.6% ± 2.3%	46.6% ± 0.9%	51.1% ± 7.9%	34.9% ± 4.9%	33.2% ± 2.1%	23.0% ± 2.5%	70.3 ± 3.8	3
APPLE-PostALA	34.0% ± 2.6%	37.8% ± 2.5%	52.7% ± 6.0%	43.0% ± 5.7%	50.9% ± 3.0%	48.7% ± 8.1%	32.7% ± 1.5%	33.0% ± 5.1%	23.5% ± 3.0%	76.0 ± 5.2	3
Local-Centralized Upper Bound	85.3% ± 2.8%	87.2% ± 2.1%	95.3% ± 0.2%	94.1% ± 0.4%	98.1% ± 0.3%	95.5% ± 0.2%	88.0% ± 2.8%	81.0% ± 4.8%	73.5% ± 4.5%	—	3

Main baselines + APPLE/FedALA/PostALA | High heterogeneity ($q=0.8$) | selector=macro_pr_auc | protocol=VV | overview bars
V+V: visible validation selection → visible test



Main baselines + APPLE/FedALA/PostALA | High heterogeneity (q=0.8) | selector=macro_pr_auc | protocol=VO
V+O: visible validation selection → oracle/full test

Method	Micro-F1	Macro Pos-F1	Micro PR-AUC	Macro PR-AUC	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1	Sel. round	n
Fully Local	46.7% ± 6.3%	44.8% ± 8.6%	46.8% ± 5.9%	48.0% ± 6.2%	46.6% ± 14.2%	48.5% ± 10.8%	46.1% ± 9.8%	44.6% ± 8.9%	38.3% ± 1.8%	—	3
FedAvg	58.1% ± 7.5%	58.8% ± 8.1%	66.1% ± 11.6%	68.1% ± 9.4%	57.9% ± 13.3%	64.6% ± 11.1%	59.9% ± 4.2%	59.8% ± 7.8%	51.7% ± 4.6%	74.7 ± 2.1	3
FedProx	60.1% ± 3.0%	60.8% ± 2.8%	73.0% ± 6.1%	75.1% ± 5.3%	61.3% ± 4.9%	66.0% ± 4.8%	61.1% ± 2.2%	62.4% ± 5.3%	53.2% ± 3.9%	78.0 ± 1.7	3
GCFL+	63.7% ± 1.8%	64.9% ± 2.8%	76.0% ± 3.4%	76.7% ± 1.0%	69.0% ± 12.0%	69.3% ± 3.1%	67.0% ± 0.9%	64.4% ± 0.6%	54.7% ± 1.5%	76.0 ± 5.3	3
FedALA-Full	60.9% ± 3.1%	61.4% ± 3.3%	67.8% ± 2.4%	69.2% ± 4.4%	63.2% ± 6.9%	65.6% ± 5.0%	63.0% ± 1.7%	61.4% ± 3.4%	54.1% ± 2.7%	78.0 ± 2.6	3
APPLE-Full	59.6% ± 4.8%	60.0% ± 5.2%	64.3% ± 5.8%	66.2% ± 7.6%	62.7% ± 11.3%	62.8% ± 6.2%	57.7% ± 3.3%	62.7% ± 7.0%	53.9% ± 1.1%	79.3 ± 1.2	3
APPLE-ALA	68.7% ± 0.8%	70.6% ± 0.5%	77.9% ± 0.2%	78.8% ± 0.8%	79.9% ± 2.2%	77.8% ± 2.4%	71.2% ± 4.5%	68.2% ± 2.3%	56.2% ± 4.2%	70.3 ± 3.8	3
APPLE-PostALA	69.9% ± 2.6%	71.8% ± 2.2%	81.4% ± 2.7%	81.9% ± 1.6%	82.0% ± 3.9%	81.2% ± 4.1%	69.4% ± 1.0%	68.7% ± 4.2%	57.7% ± 4.6%	76.0 ± 5.2	3
Local-Centralized Upper Bound	85.3% ± 3.1%	87.3% ± 2.5%	95.6% ± 0.3%	94.2% ± 0.7%	98.9% ± 1.4%	94.8% ± 0.5%	88.1% ± 2.5%	81.9% ± 5.7%	72.8% ± 4.6%	—	3

Main baselines + APPLE/FedALA/PostALA | High heterogeneity ($q=0.8$) | selector=macro_pr_auc | protocol=VO | overview bars
V+O: visible validation selection → oracle/full test

